

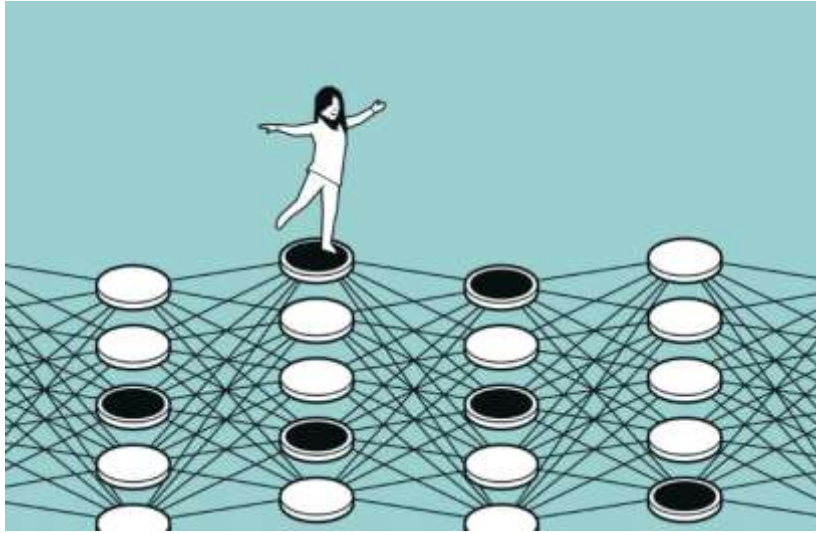
A. Ghasemifard, S. Zhao, A. Ugarte, D. Dzujah, E. Dongmo, P. Bhauriyal, S. Das, and M. Hecker

Chair of Theoretical Chemistry at Technical University of Dresden and CASUS

Machine Learning For Interatomic Potentials And Materials Discovery

Bad Schandau Retreat Tuesday, 25 March 2025 at 8:30

Nobel Prize in Physics and Chemistry 2024



- They used physics to find patterns in information



John J. Hopfield



Geoffrey Hinton



- They cracked the code for proteins' amazing structures



David Baker



Demis Hassabis



John Jumper

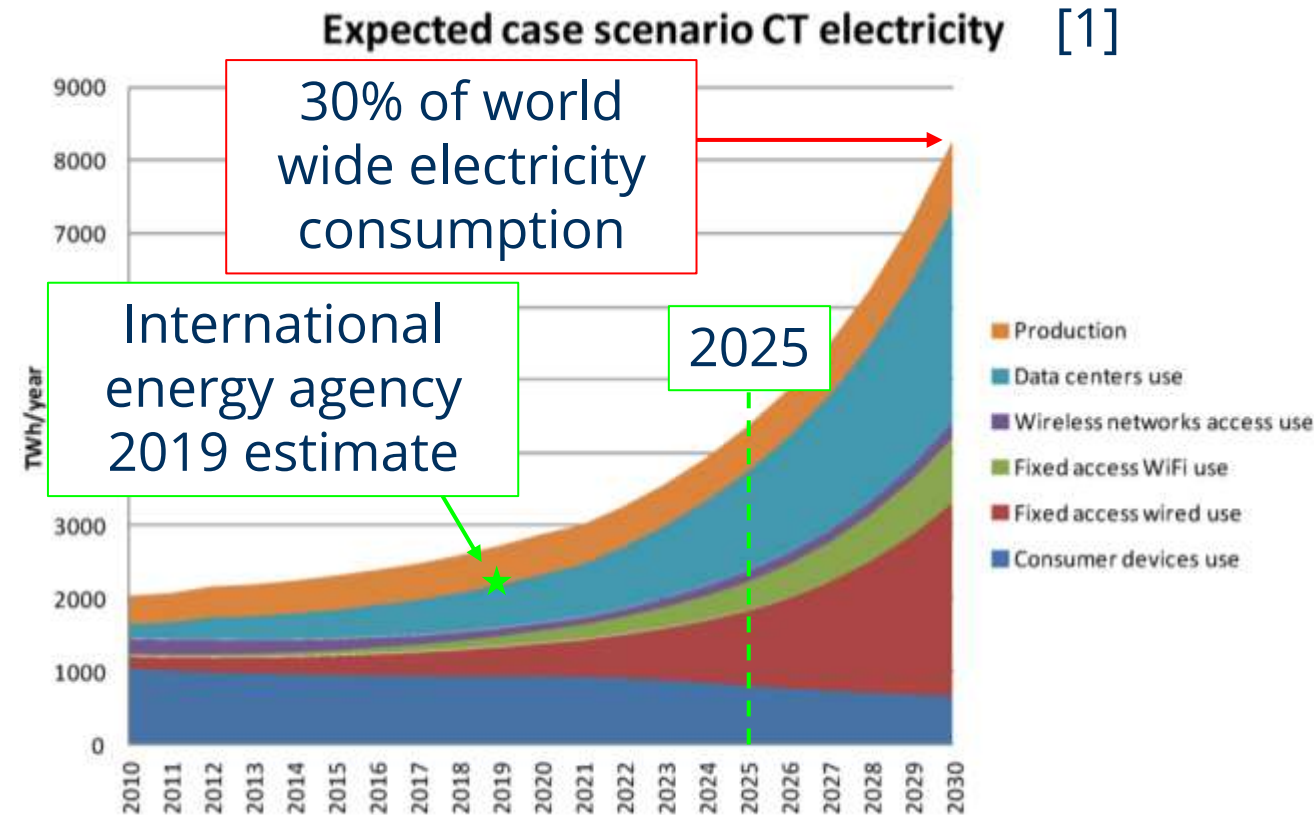
ICT Massive Energy Consumption

Energy consumption from **I**nformation and **C**ommunications **T**echnologies increasing rapidly

Large portion: data centers, will only grow faster due to

- 1) HPCs
- 2) Clouds
- 3) AI

→ Need more energy-efficient hardware/software



Machine-learning Session



Alireza

Shuangjie

- Introduction to machine-learning interatom potentials (MLIP)
- Why machine learning at all?
- The basics of neural networks (NN)



7+4 slides



6+7 slides

- MLIP challenges → Long-range interactions
- Selected/popular MLIPs and their application in materials science



Adriana

Diyan

Machine-learning Session



Elvira

Preeti

- Introduction to AI-driven materials discovery
- Machine-learning for property prediction

3

6+8 slides

4

6+10 slides

- Inverse design of materials
- Case studies of materials design
1. MatterGen 2.GNoME



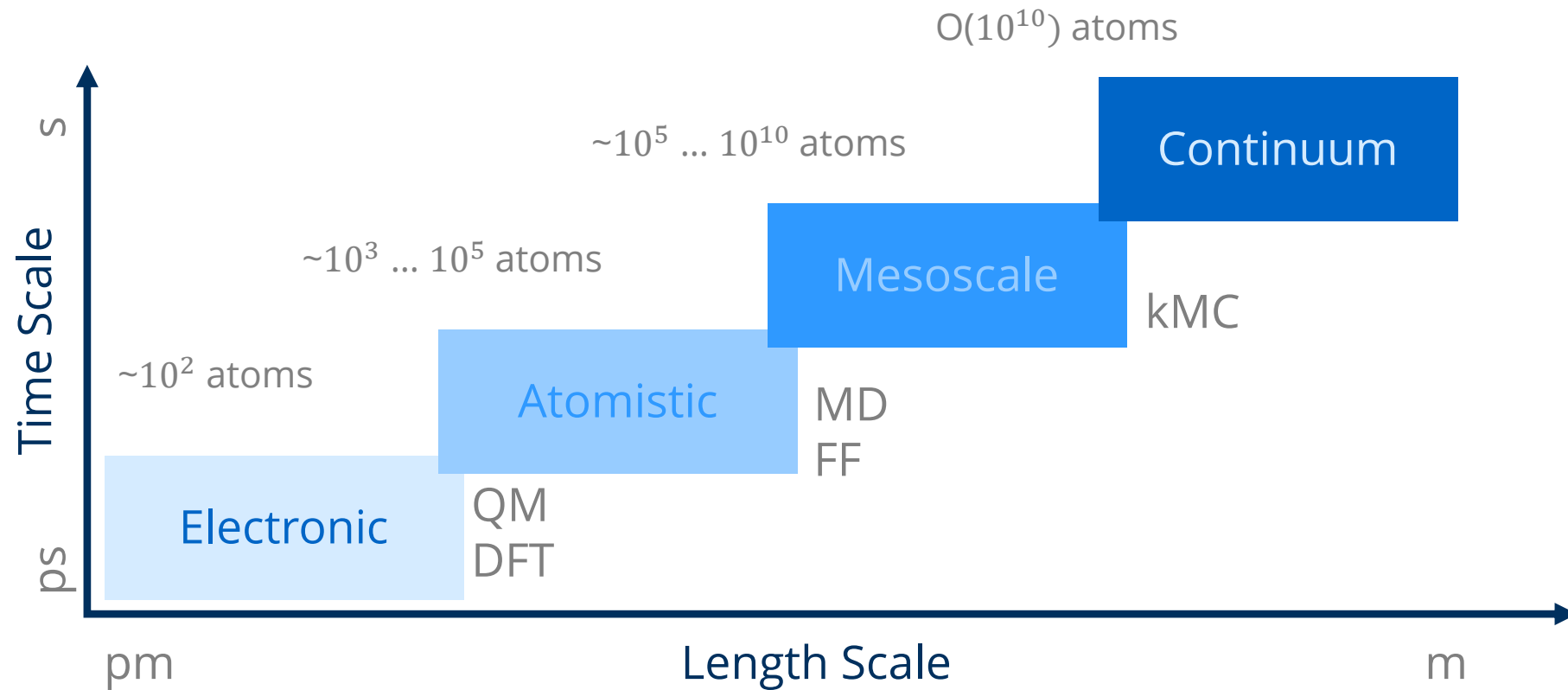
Shubhajit **Maria**



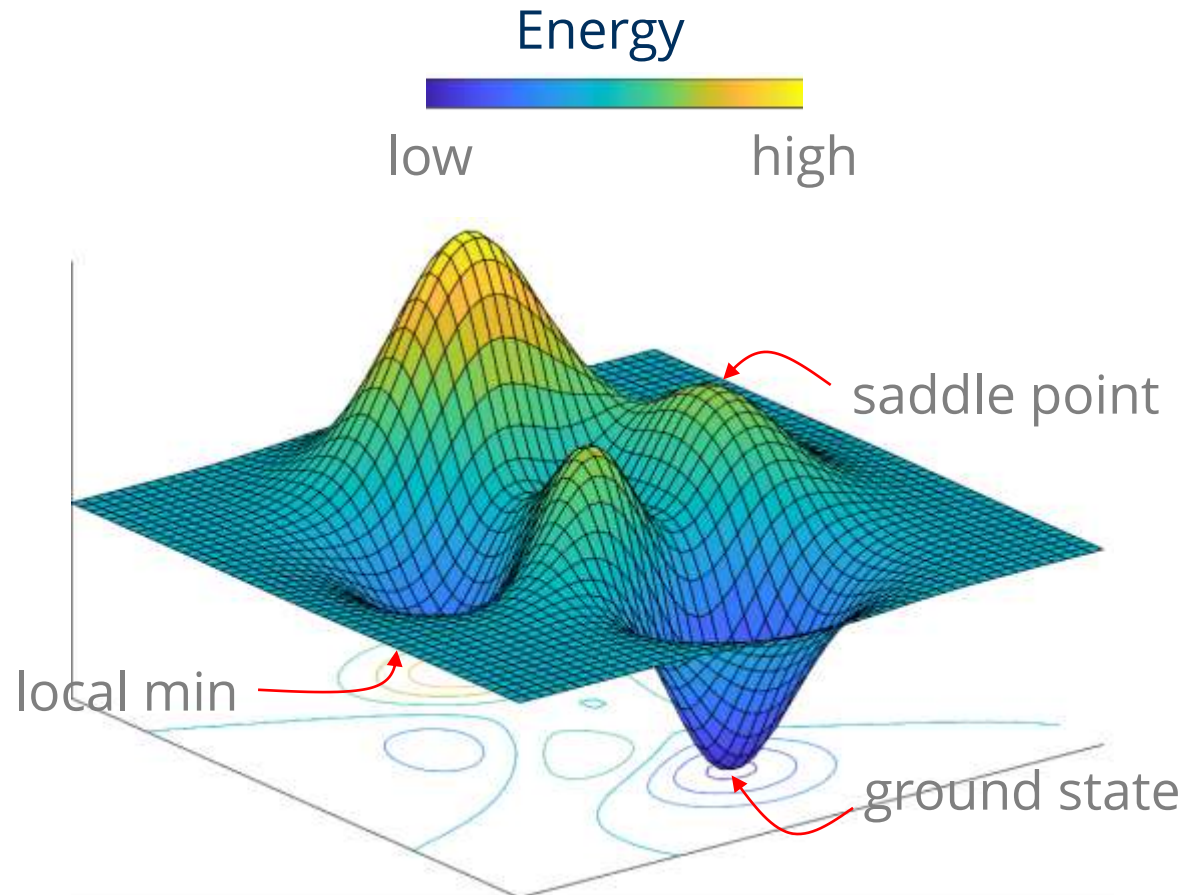
Ali

1. Introduction

Materials simulations span multiple length and time scales



Potential Energy Surface (PES)



In the atomistic world, several properties of interest are obtained from a PES

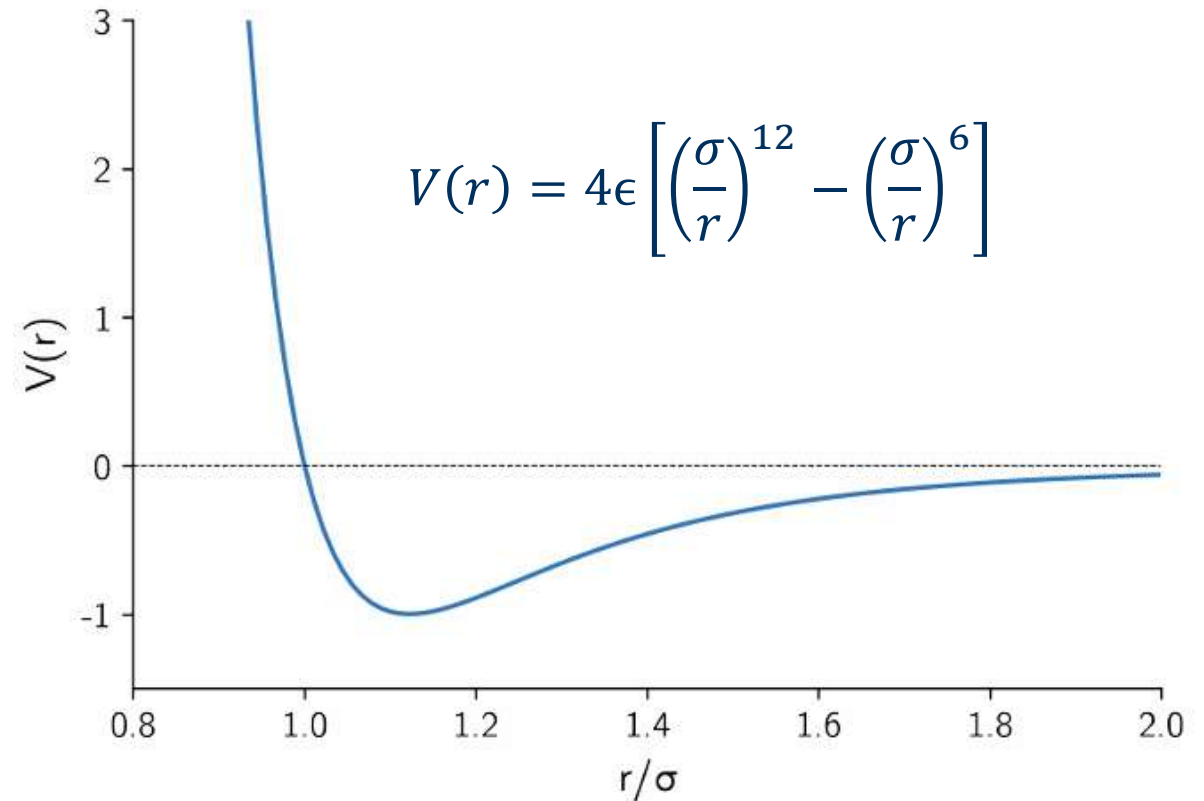
- From a PES, we can derive: geometries, reaction energies, mechanical properties, dynamical behavior...
- A PES is illustrated with respect to system coordinates (positions, distances, angles etc.)
- However, in principle, the energy from atomistic systems can be computed from the atomic numbers and coordinates:

$$E = f(Z_i, r_i)$$

Simple Model Interatomic Interactions

Lennard-Jones potential: a simple example of pair potential

- Instead of using quantum mechanics or DFT to model a PES, we can use simpler models to describe interatomic interactions.
- This reduces the computational cost: analytical energy functions are much faster than quantum mechanics calculations (several orders of magnitude).



Complicated Systems = Several Potentials

- Lennard-Jones potential

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

- Morse potential

$$V(r) = A[1 - e^{-a(r-r_{eq})}]$$

- Harmonic potential

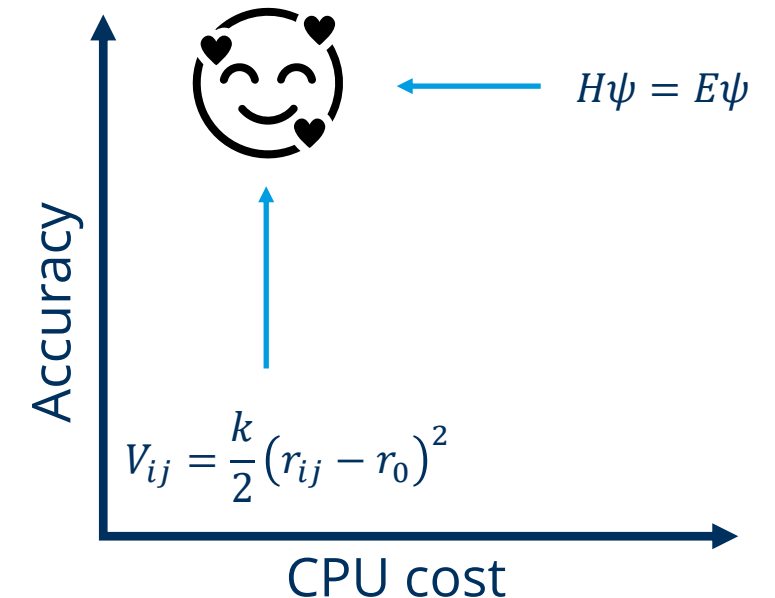
$$V(r) = A(r - r_{eq})^2$$

- Buckingham-Coulomb potential

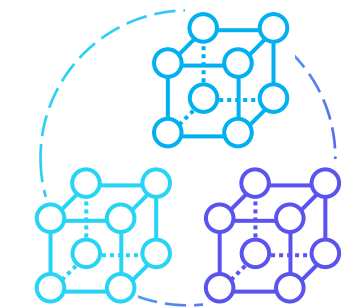
$$V(r) = Ae^{-Br} - \frac{C}{r^6} + \frac{q_1 q_2}{4\pi\epsilon_0 r}$$

Problems:

- How to choose the functional forms and parameters?
- how can we obtain accurate, yet fast potentials?



MLIP Workflow

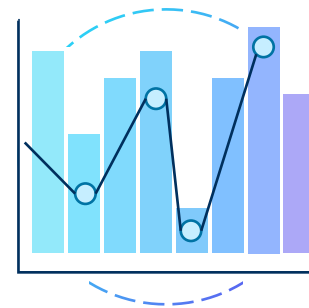


Generate initial structures



Compute training data:

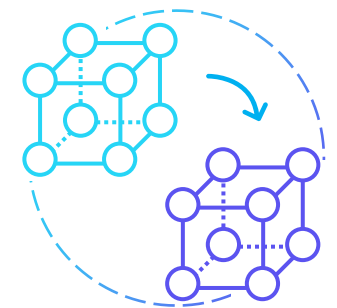
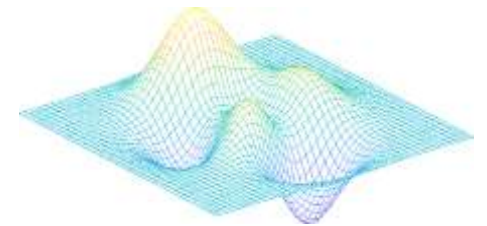
- Energies, forces: DFT



Fit to the training data



Preliminary MLIP

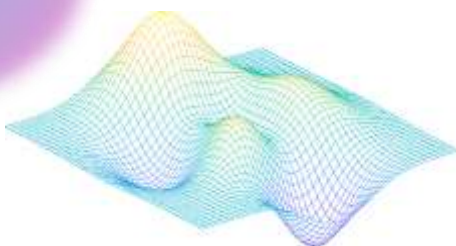


Run MLIP to get result



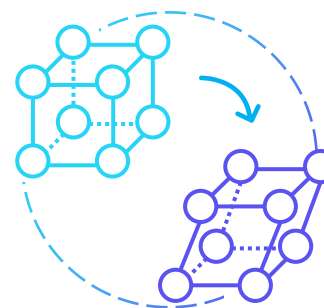
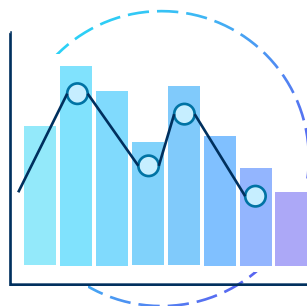
1. Compute additional training data with DFT

Active learning

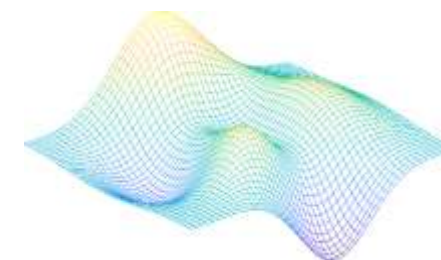


3. Improved MLIP

2. Fit to the training data



Run MLIP to get result



MLIP

ready to use

Challenges

- 1 Selecting the most suitable potential for accurate modeling

**Potential
Choice**

- 2 Ensuring potential applicability across different systems

**Transferability
Issue**

- 3 Addressing challenges in modeling long-range interactions

**Long-range
Interactions**

- 4 Developing potentials for accurate magnetic systems

**Magnetic
Systems**

**Improved
Interatomic
Potentials**



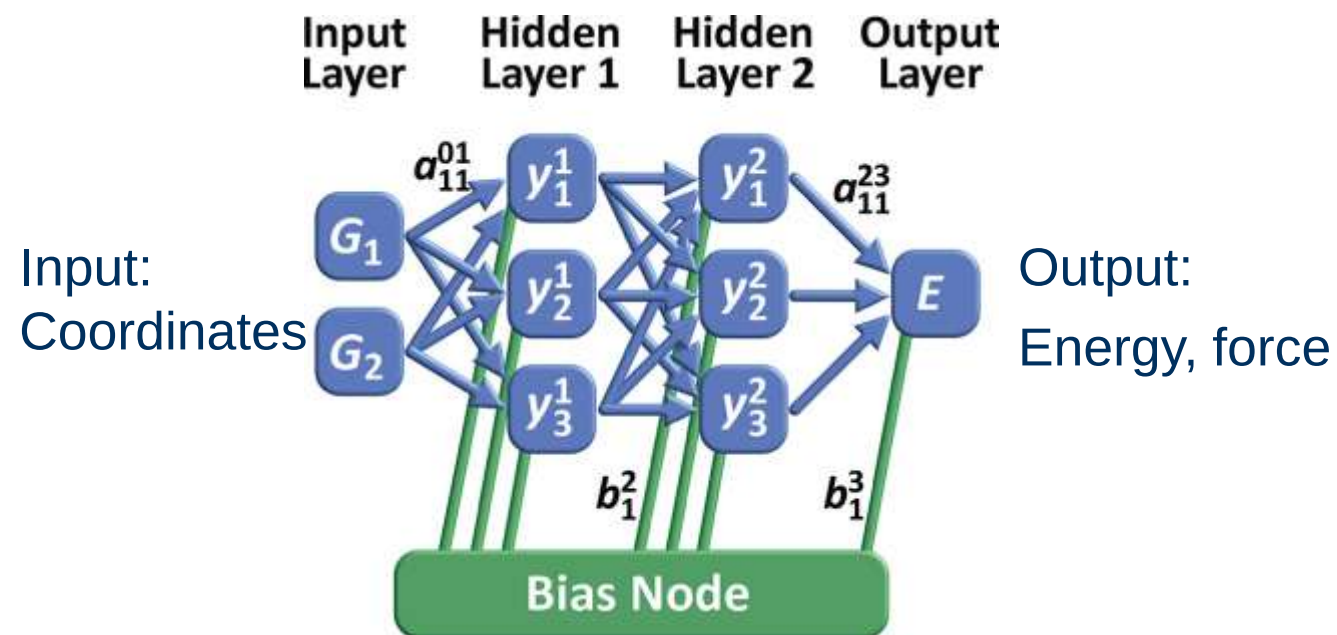


Shuangjie

2. NN Interatomic Potentials

Neural Network Potentials: The Basics

1. Generation
Low-Dimensional Systems
global description



High-Dimensional Systems

short-range

long-range

non-local

2. Generation

local atomic energies

yes

no

no

3. Generation

local atomic energies and local charges

yes

yes

no

4. Generation

local atomic energies and global charges

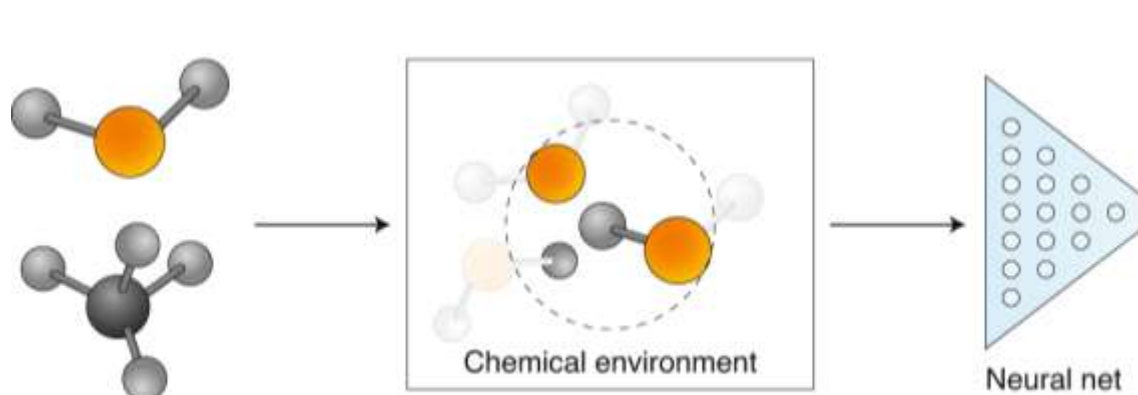
yes

yes

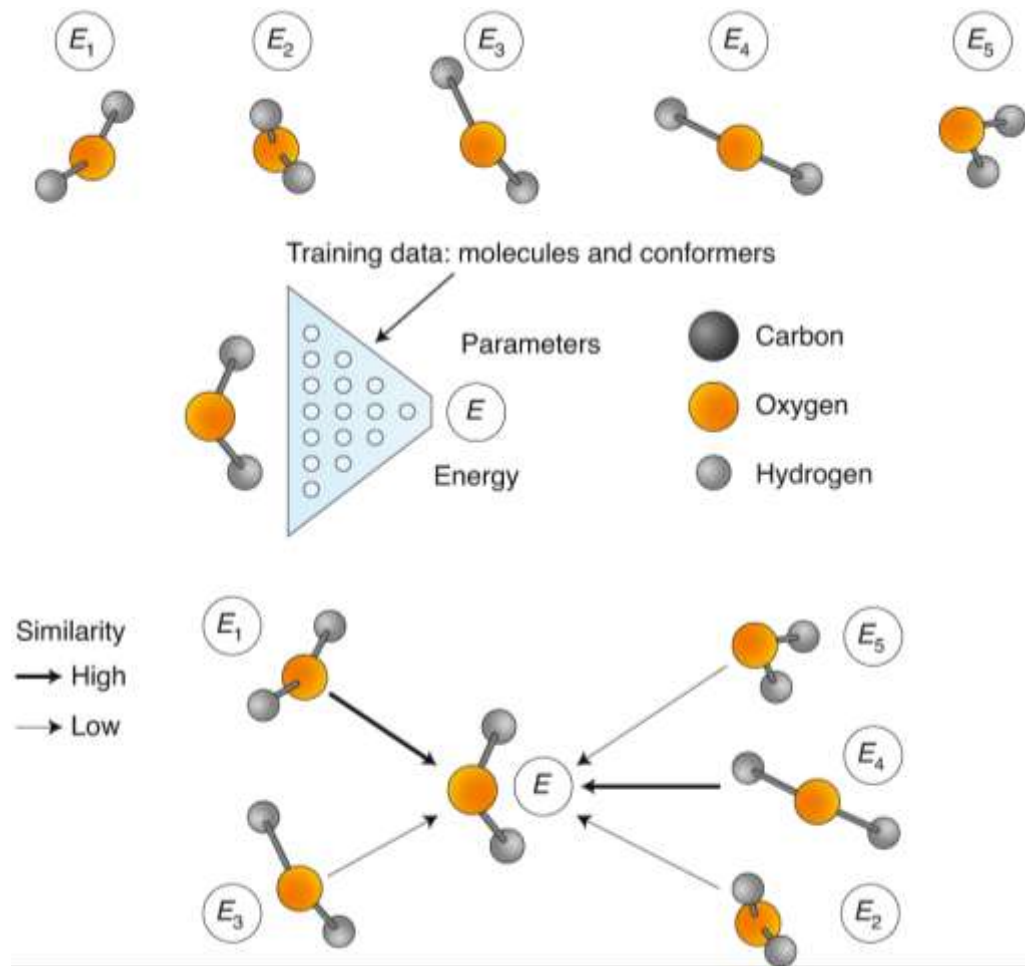
yes

Feature Representation (Descriptor)

What is and how to find a representation?

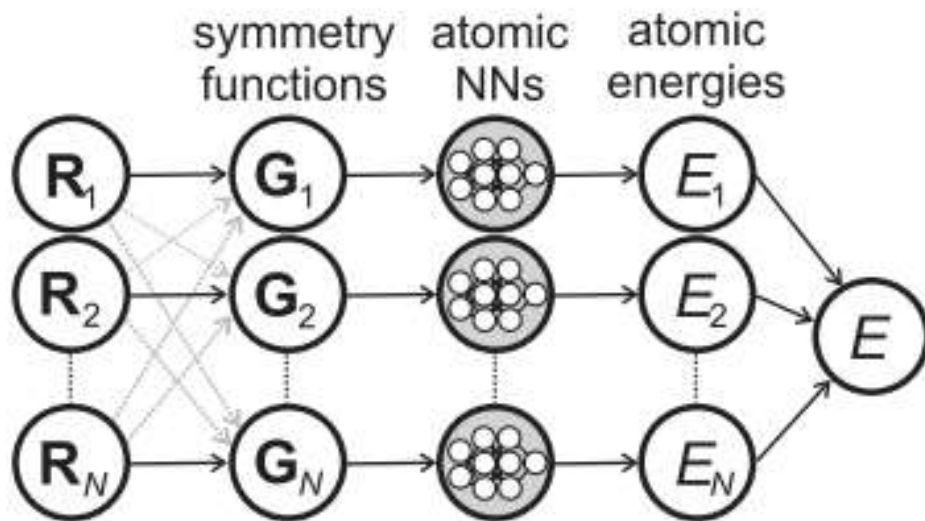


The challenge of finding a fixed-size representation of a system of arbitrary size can be solved using representations of atom centered local chemical environments.



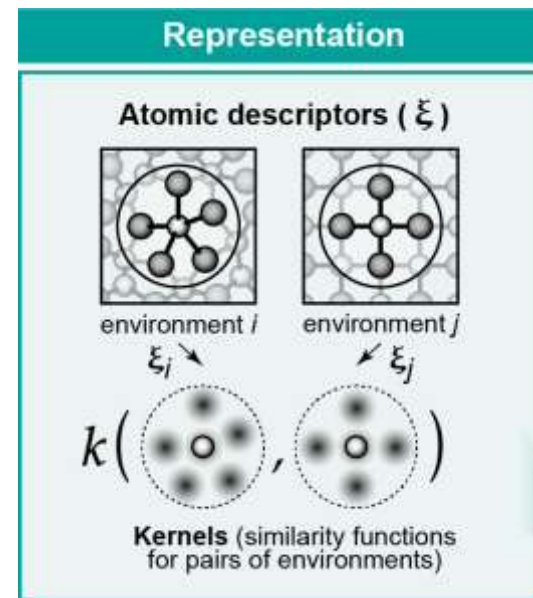
Feature Representation (Descriptor)

Symmetry Functions



To avoid getting different energy due to a translation or rotation

SOAP-GAP

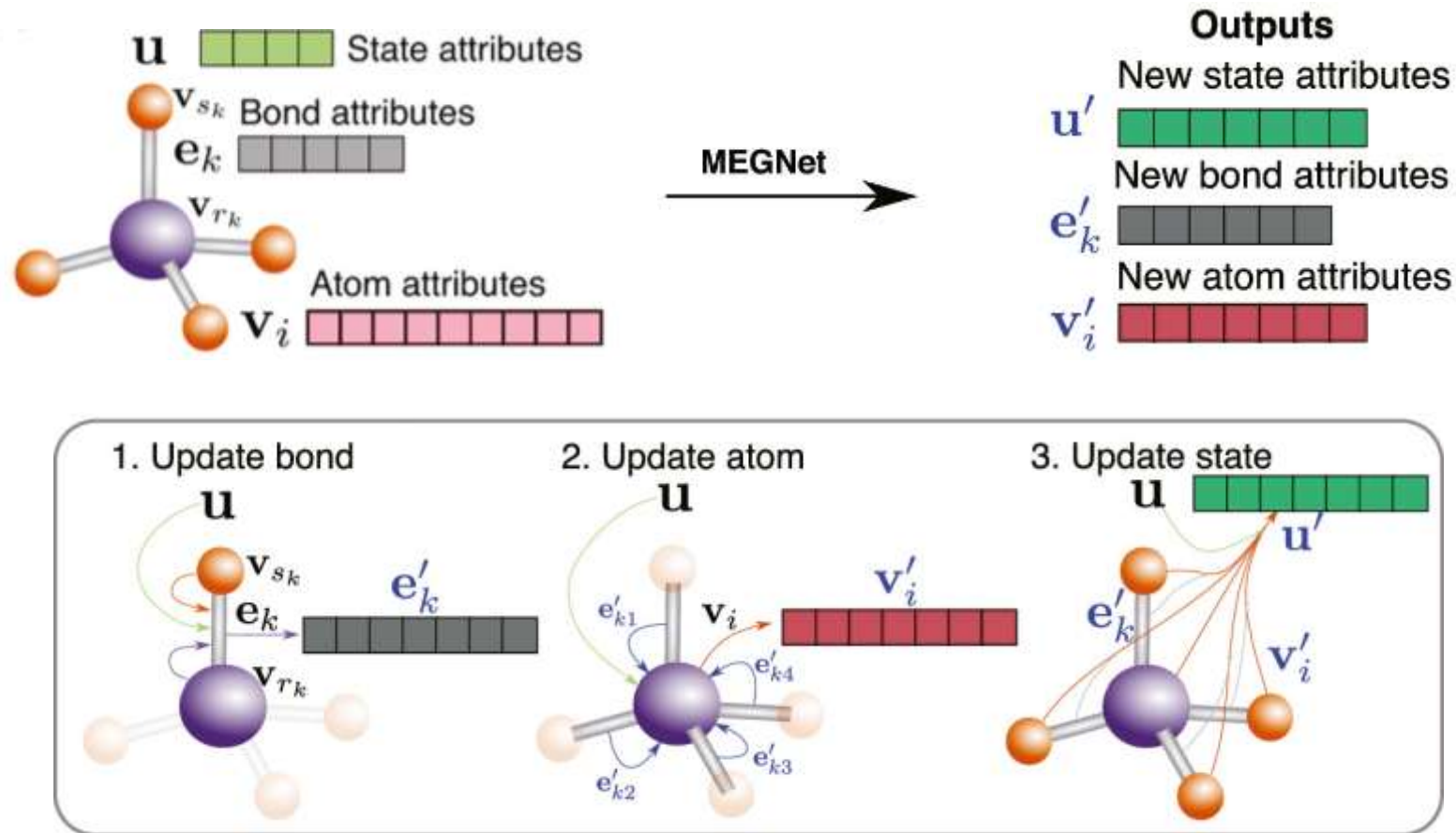


Smooth Overlap of Atomic Position used to Gaussian Approximation Potential.

4. Jörg Behler. J. Chem. Phys. 134, 074106 (2011).

5. Volker L. Deringer, Albert P. Bartók, Noam Bernstein, David M. Wilkins, Michele Ceriotti, and Gábor Csányi. *Chemical Reviews* **2021** 121 (16), 10073-10141

Graph Neural Networks (GNN)



Examples: CGCNN, MEGNet, SchNet, PhysNet ...

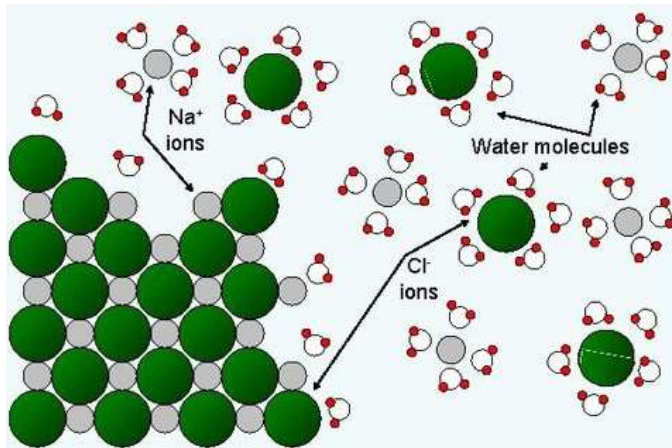


Adriana

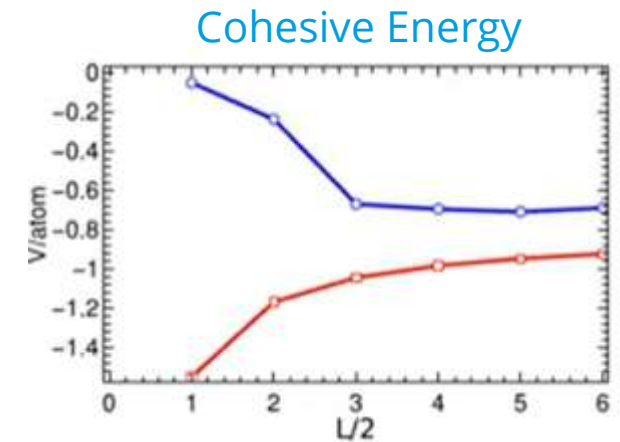
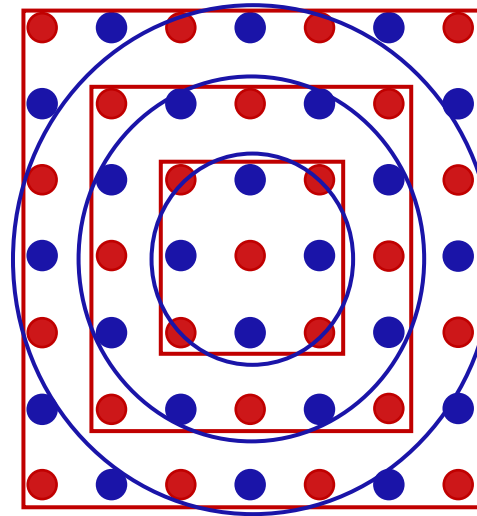
3. Interatomic Potentials and Long-range Interactions

The Challenge of Long-Range Potential Interactions

“A long-range potential is one whose range, the distance of effective influence, is unbounded or infinite.”

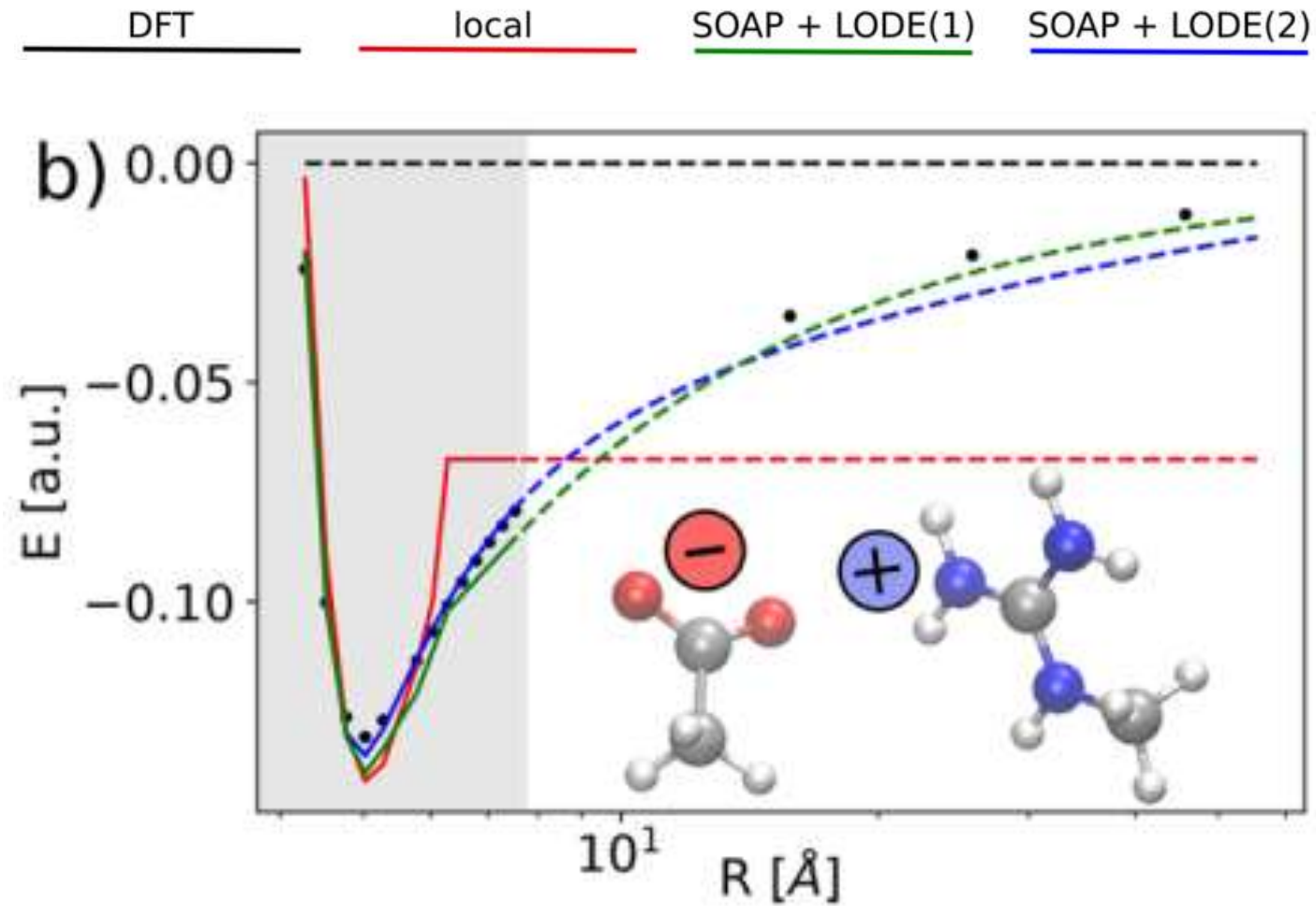


Charge transfer and hydration effects extend beyond local cutoff

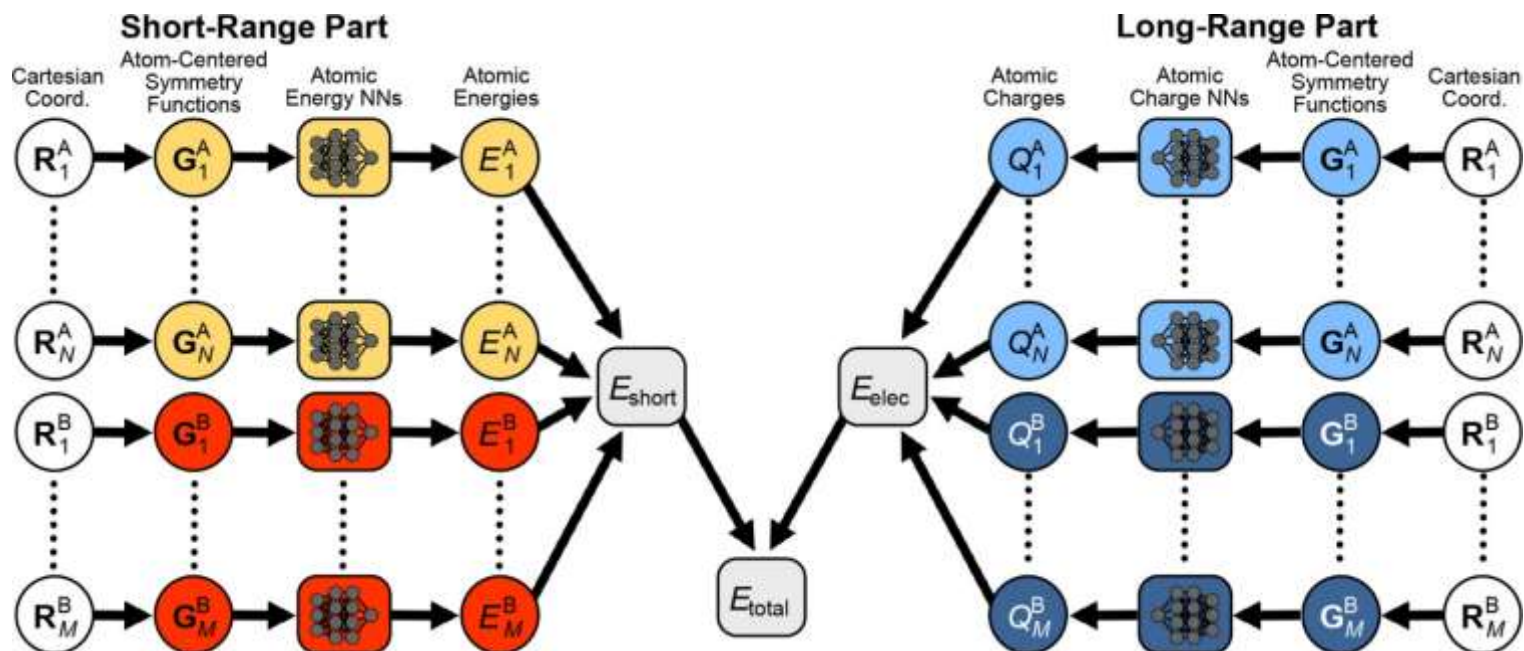
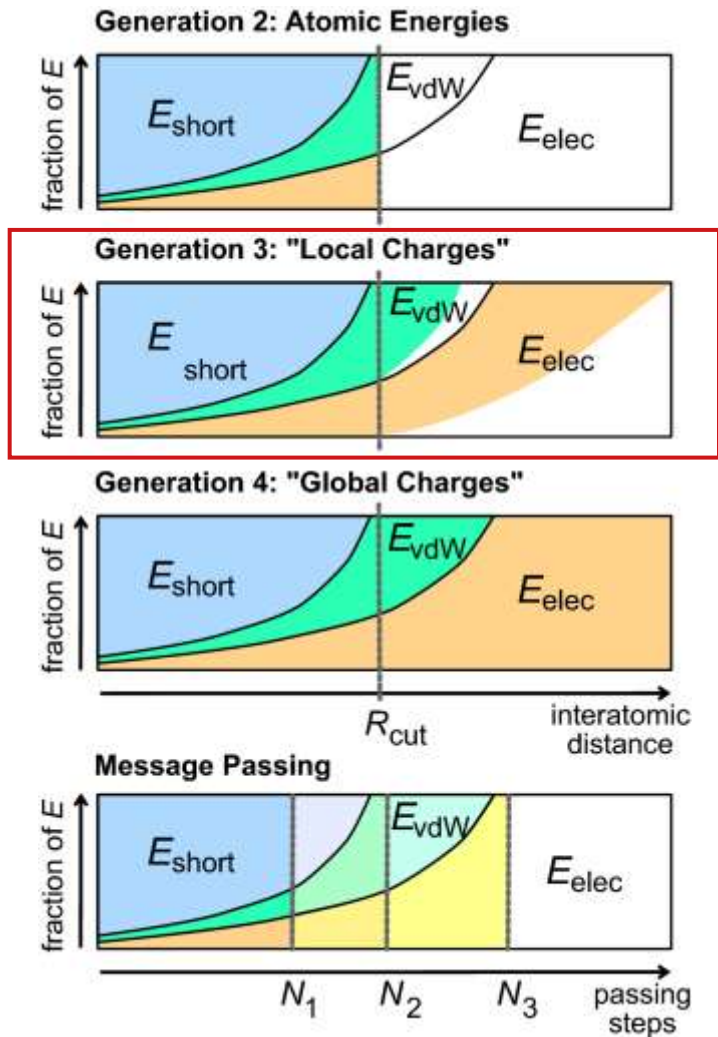


Traditional ML models rely on locality, but electrostatic interactions follow a $1/r$ dependence

The *Cutoff* Problem

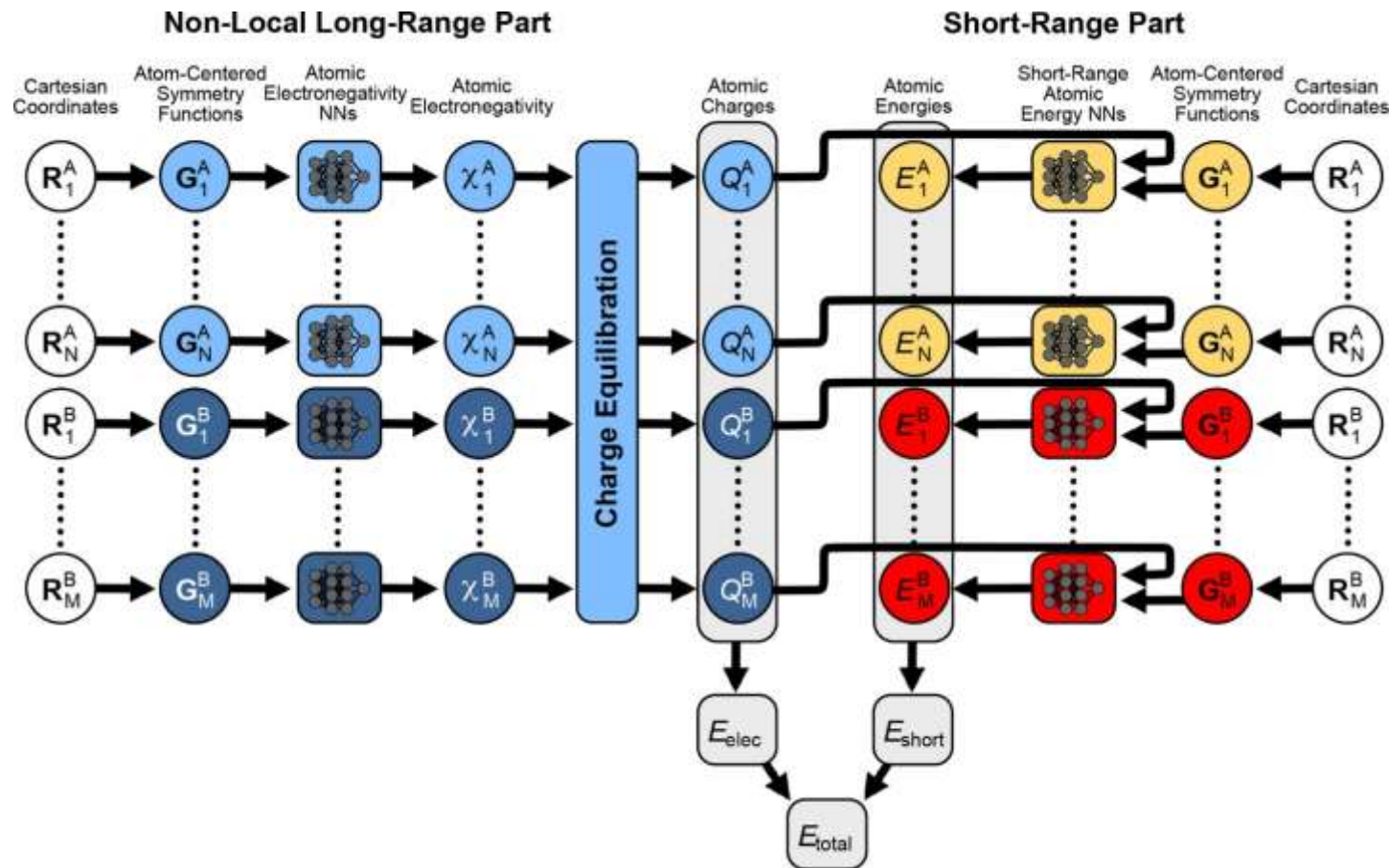
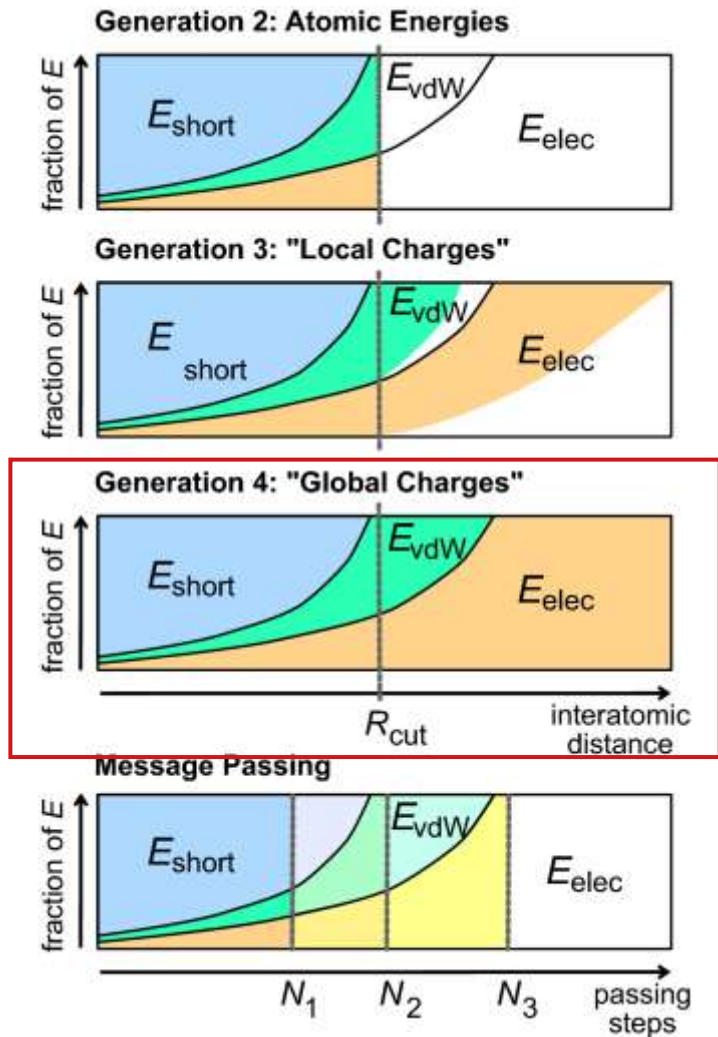


Beyond Locality



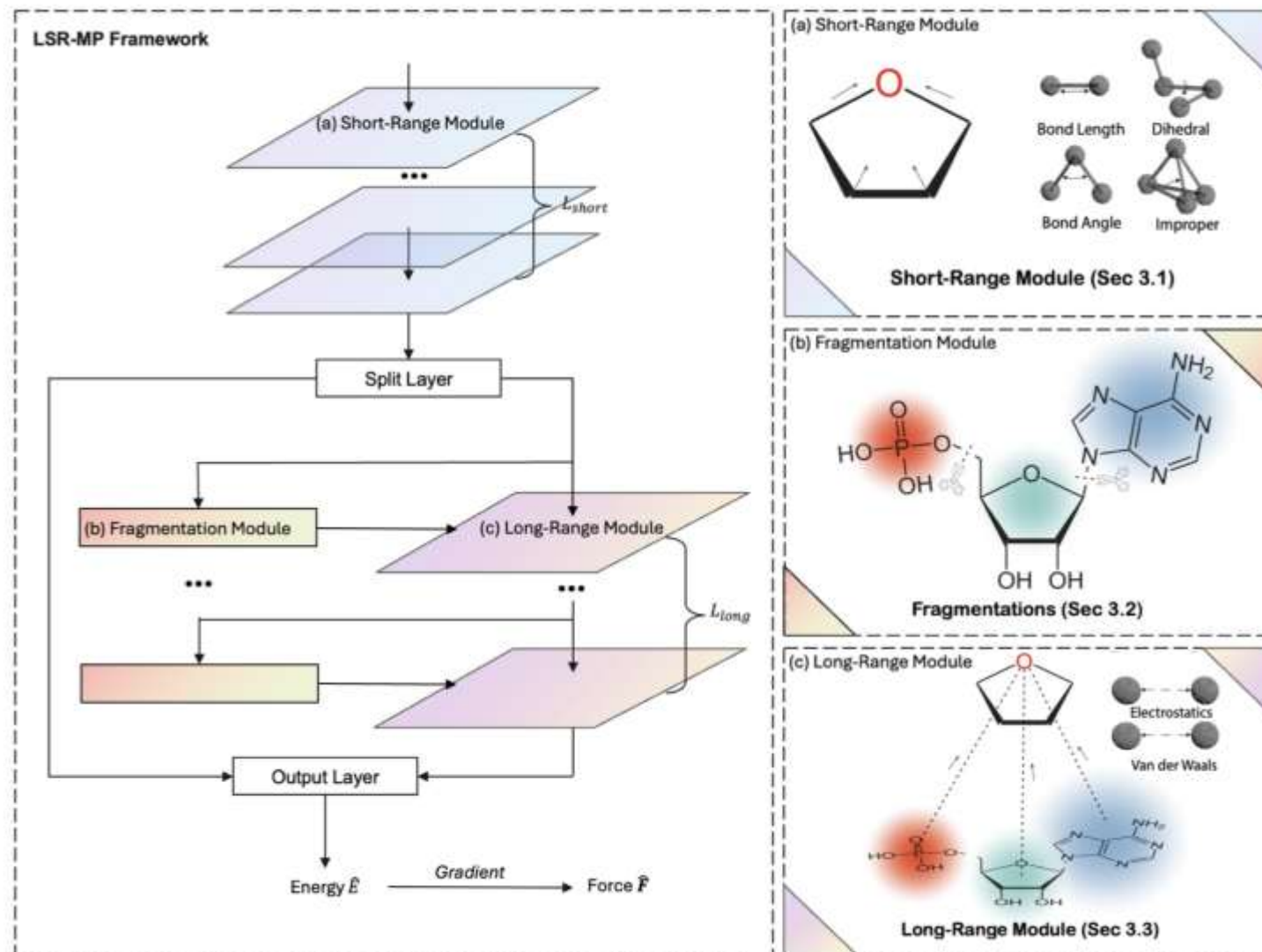
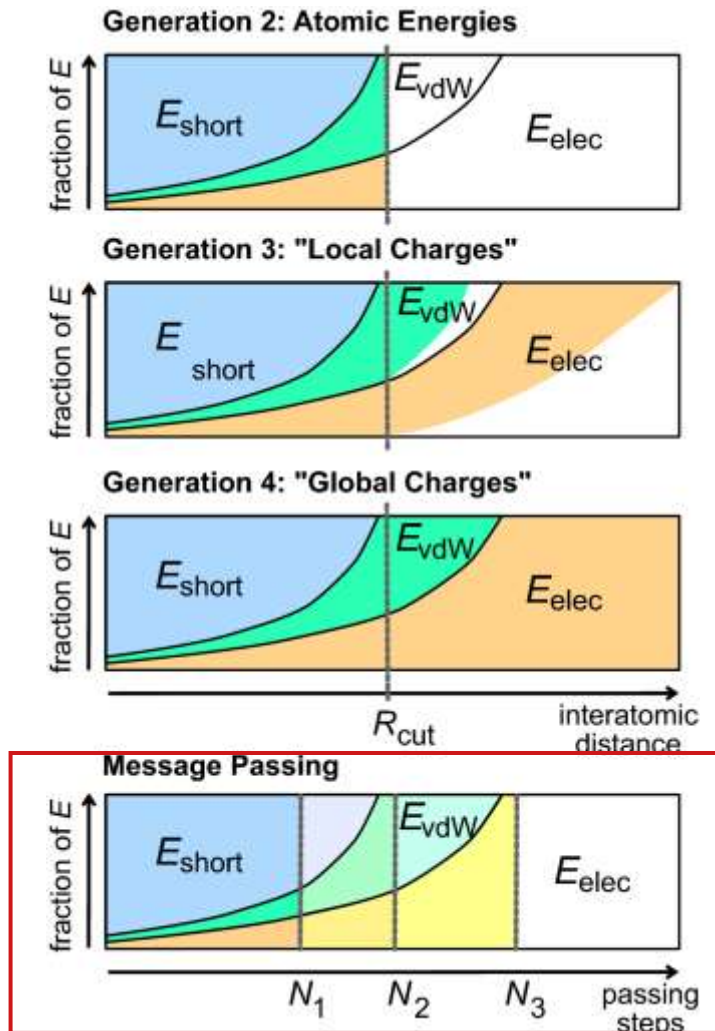
Adds localized charge models but fails for non-local effects

Beyond Locality



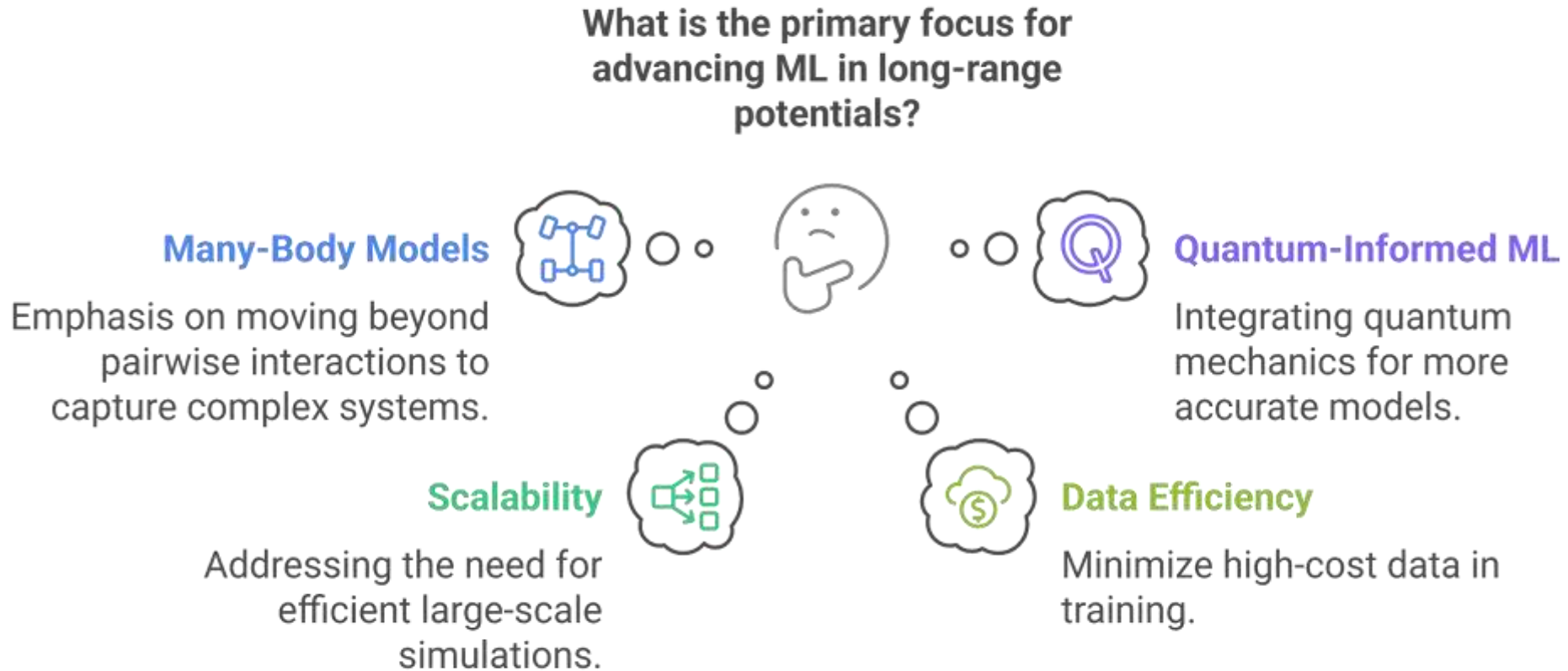
Introduces global charge equilibration for improved electrostatics

Beyond Locality



Neural networks learn relationships between distant atoms dynamically.

Future Perspectives





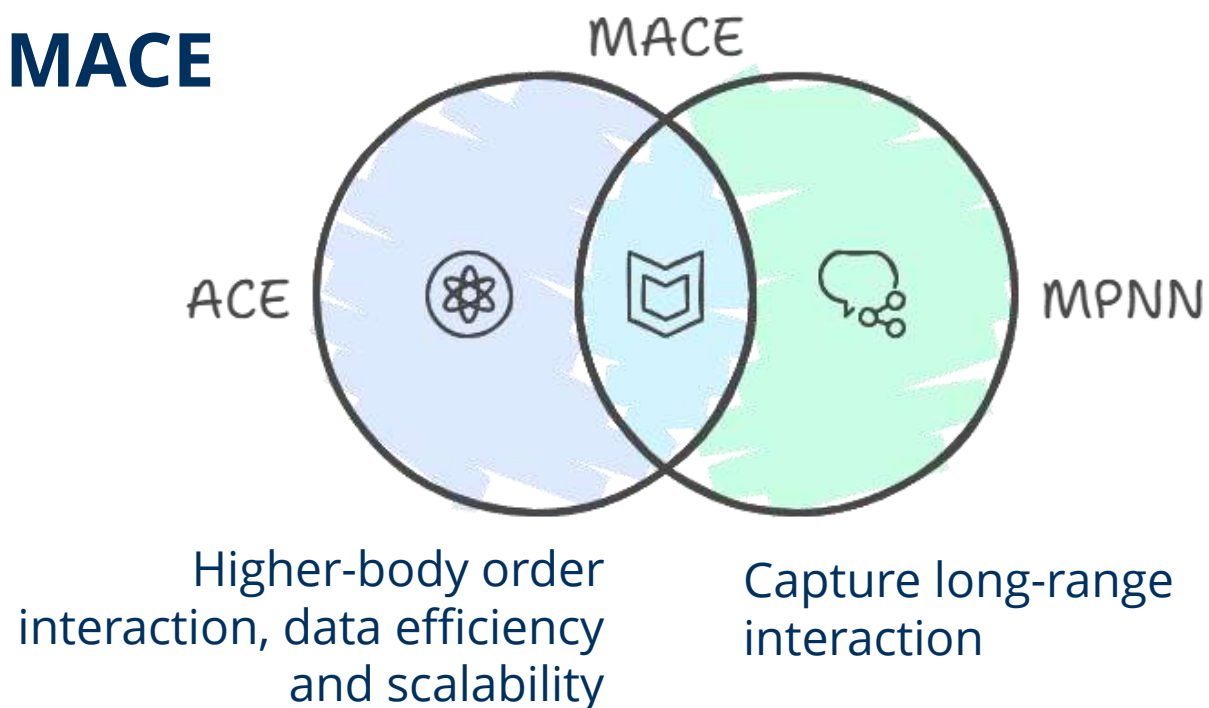
Diyan

4. Popular MLIP Models And Their Applications in Material Science

Application of Machine Learning Interatomic Potential

- 1 MACE (Multi Atomic Cluster Expansion)
- 2 CHGNET (Charge Neural Network)
- 3 M3GNET (Materials Graph Neural Network with Three-Body Interactions)

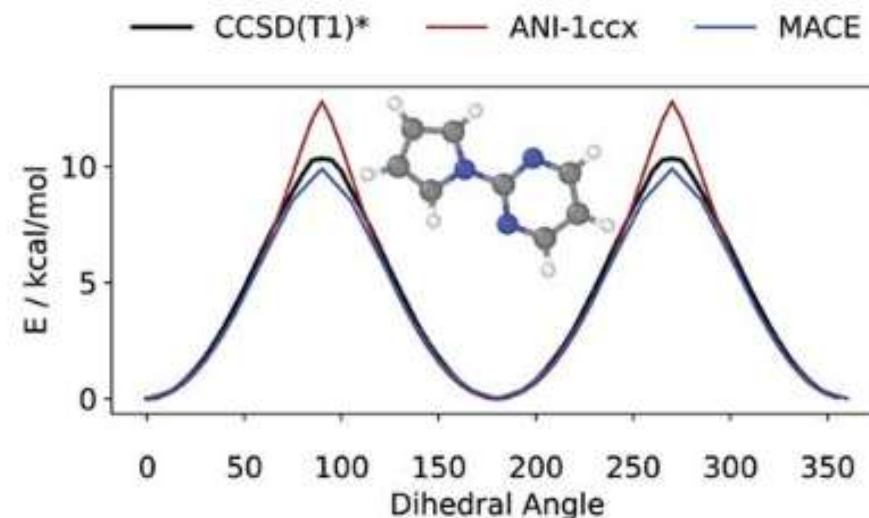
MACE



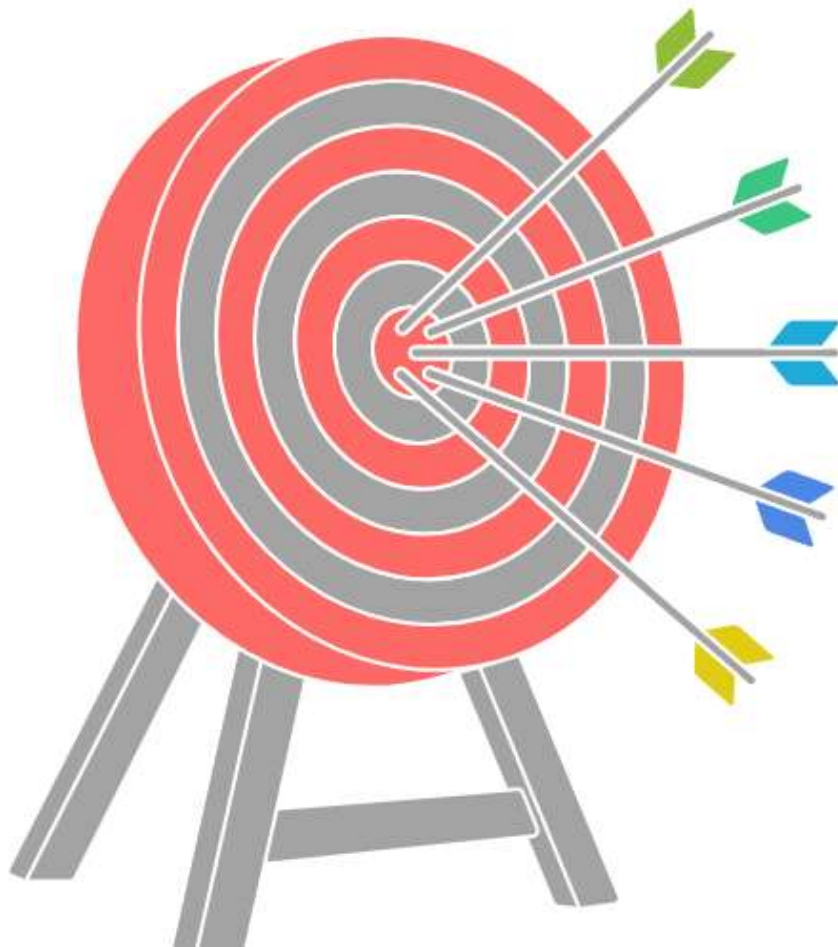
Best for:

- High-accuracy molecular dynamics simulations
- Material property prediction
- Structure optimization

Dihedral angle calculation with different methods



CHGNET



Advantages:



Charge-Informed Modeling

Includes magnetic moments for atomic charge states



GNN Architecture

Graph representation of atomic structures



Pretraining on DFT Data

Trained on large-scale DFT dataset for accuracy



Captures Electronic & Ionic Interactions

Modeling complex interactions in materials



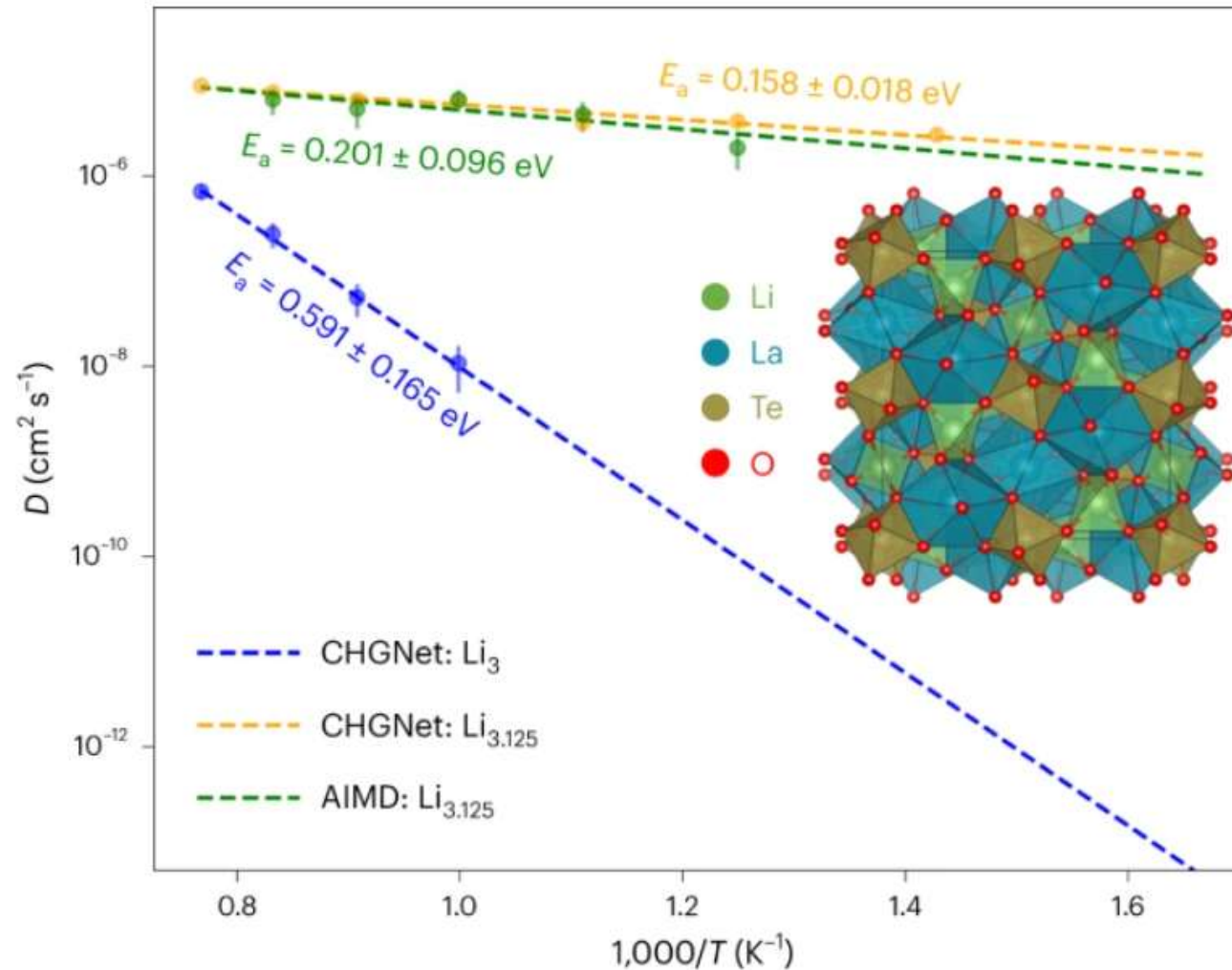
Efficient and Scalable MD

Enabling large-scale simulations



CHGNET

Li diffusivity in garnet $\text{Li}_3\text{La}_3\text{Te}_2\text{O}_{12}$



Best for:

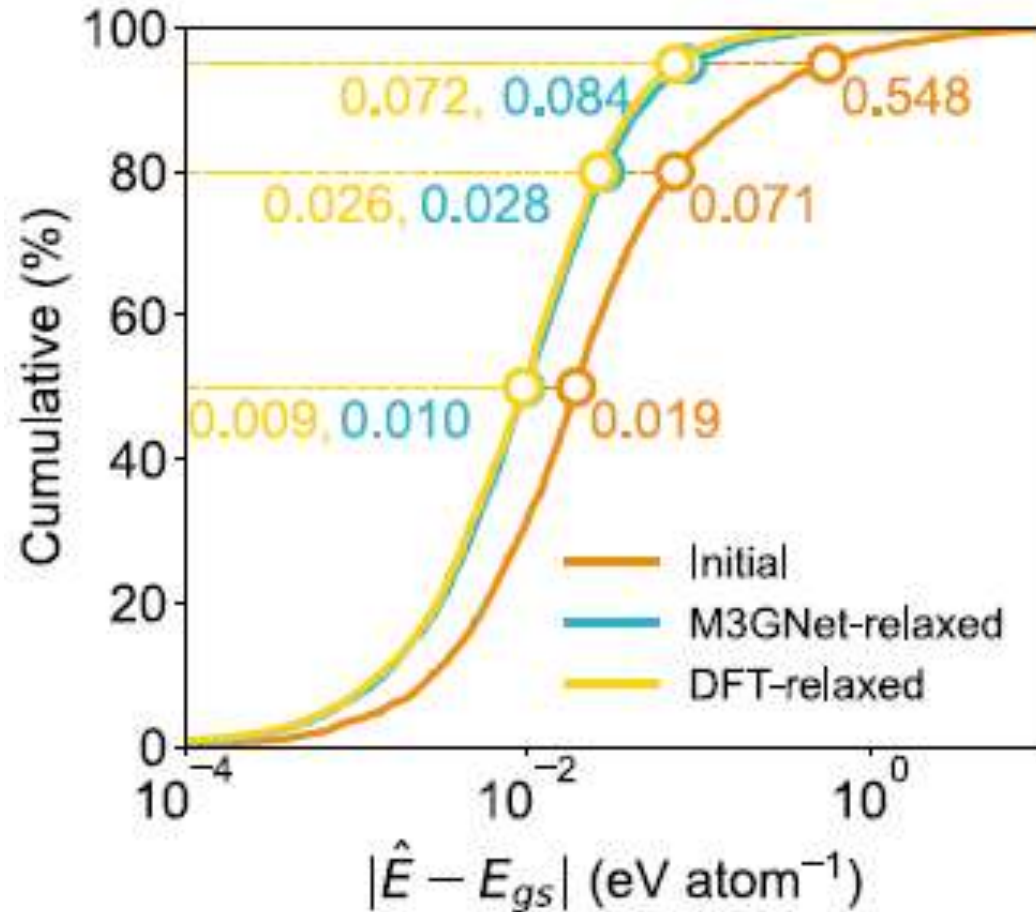
- Modeling charge transfer in materials
- Battery materials and ionic conductors
- Electrochemical interface studies

M3GNet



M3GNet

Relaxation of crystal structure with M3GNET
for the test dataset of 3,140 materials



Best for:

- Crystal structure prediction and relaxation
- Large-scale material screening

Summary

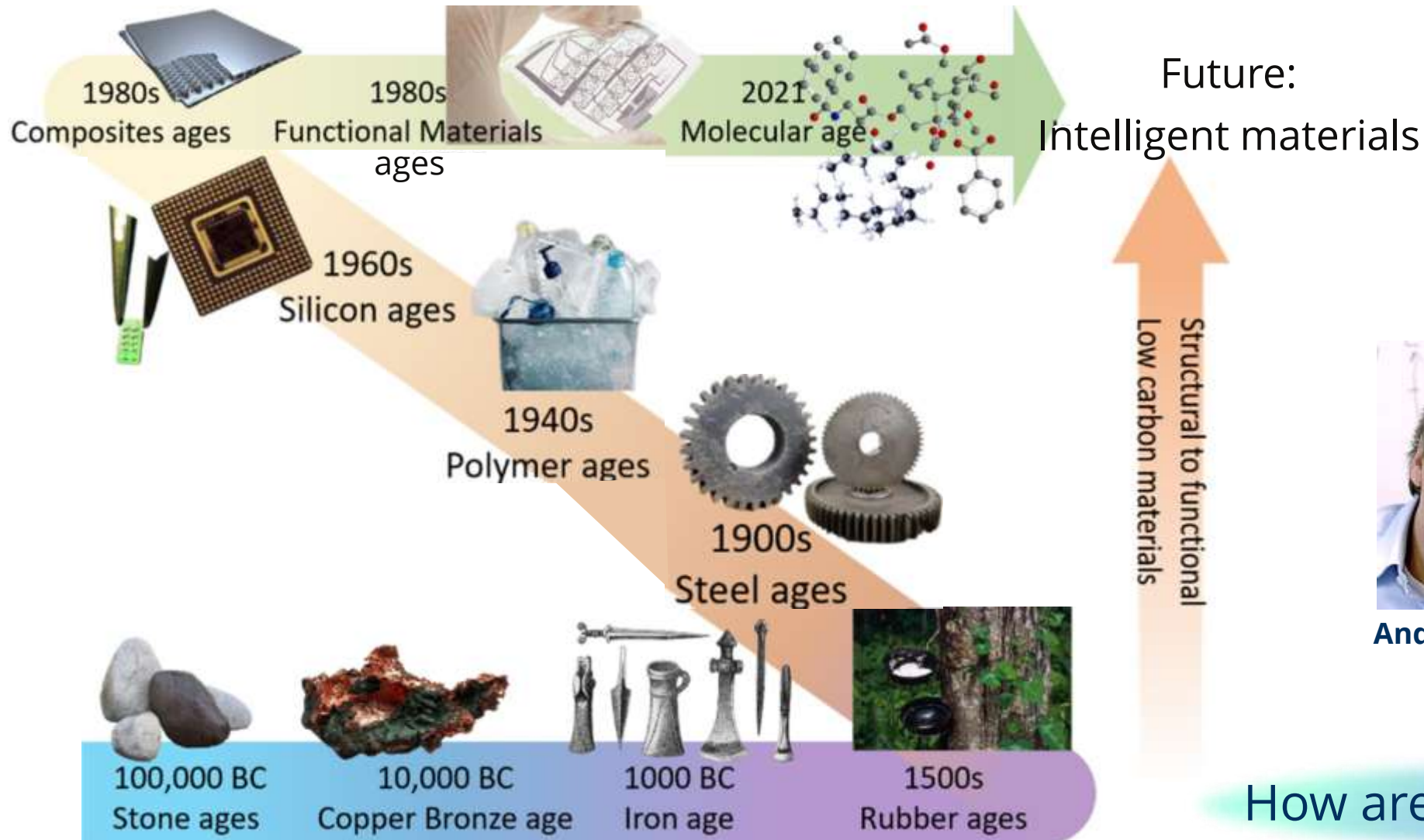
Feature	M3GNet	MACE	CHGNet
Scope	Universal IAP for all periodic table elements	Molecular dynamics	Charge-sensitive materials
Charge Sensitivity	✗	✗	✓
Best For	Materials discovery, phase stability, general IAP	Energy and force predictions for solid-state and molecular systems	Battery materials, transition metal oxides, charge-dependent systems



Elvira

5. Introduction to AI-driven materials discovery

Materials Discovery Precursor to Progress in Society



Discovery of Graphene



Andre Geim

Konstantin Novoselov

How are materials discovered?

Exploring Databases

Which database is specifically dedicated to storing **crystal structures** of discovered materials?

A NASA Database

C NOMAD

B ICSD (Inorganic Crystal Structure Database)

D Facebook Materials Index



Exploring Databases

Which database is specifically dedicated to storing **crystal structures** of discovered materials?

A

NASA Database

C

NOMAD

B

ICSD (Inorganic Crystal
Structure Database)

D

Facebook Materials
Index



How are materials discovered?



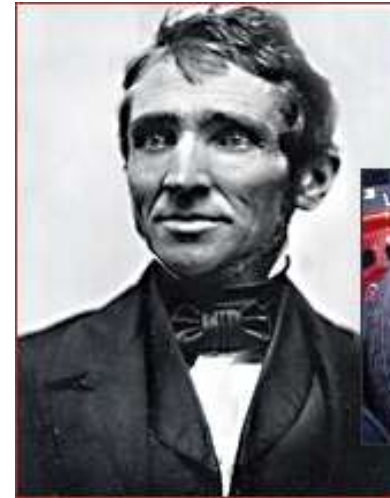
Luck / Mistakes



Persistent, long-term experiment



**Trial-and-error
Edisonian approach**



**Charles
Goodyear**



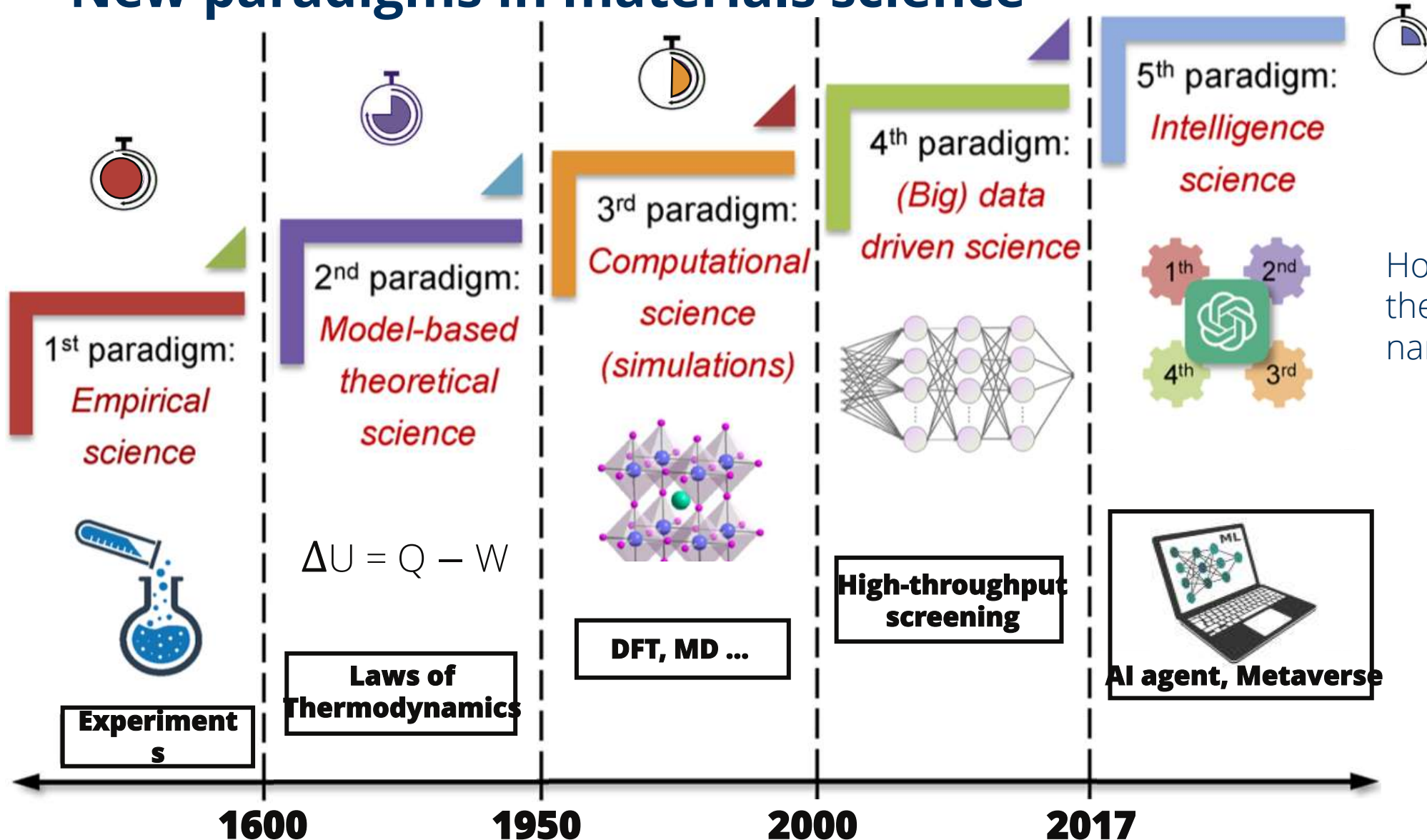
Invented vulcanized rubber in 1839

Thomas Edison



Invented the practical, long-lasting electric light bulb in 1879

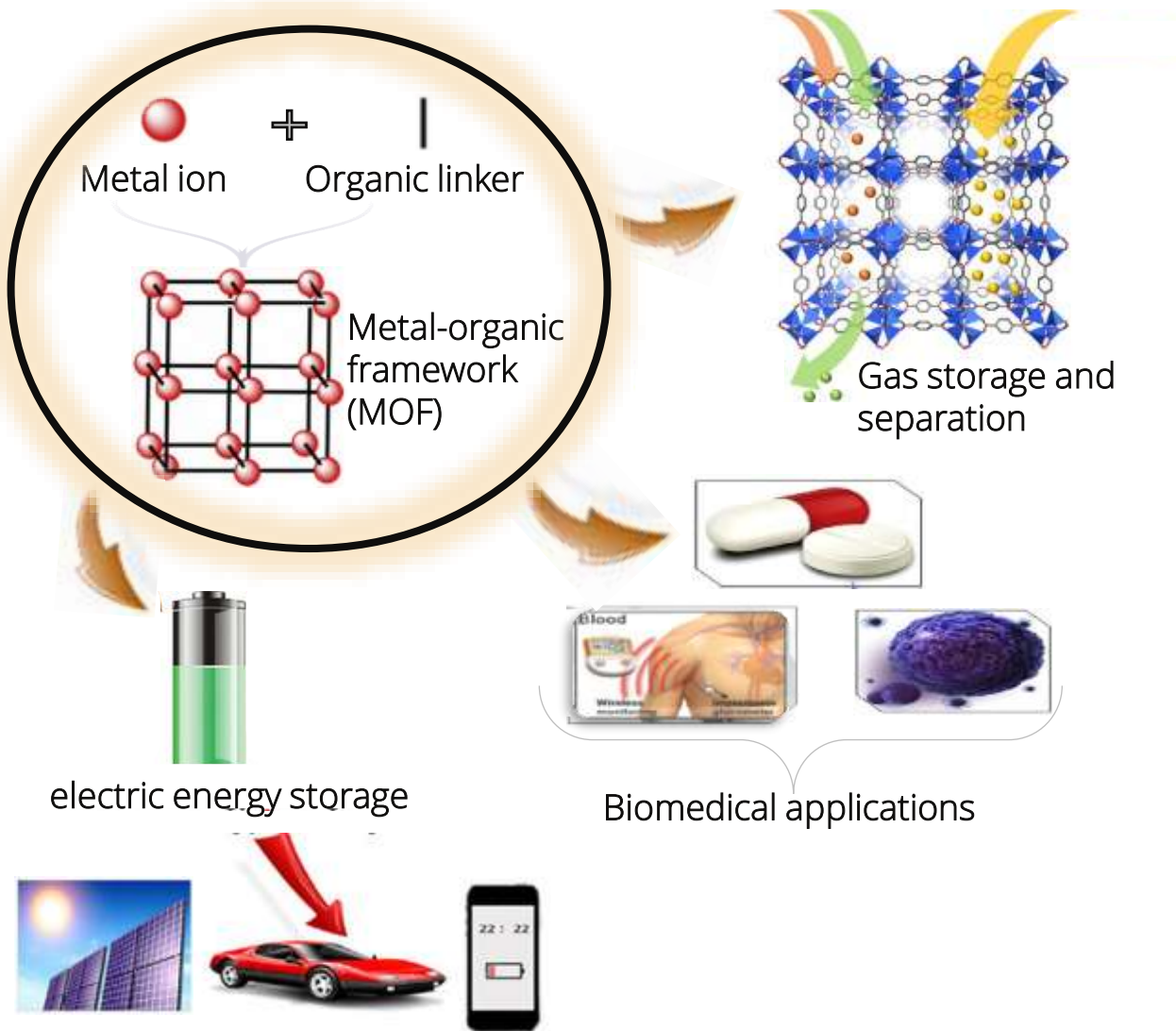
New paradigms in materials science



How can AI improve the discovery of nanoporous materials?



Designing novel Nanoporous Materials: MOF



pubs.acs.org/Langmuir

Metal–Organic Frameworks through the Lens of Artificial Intelligence: A Comprehensive Review

Kevizali Neikha and Amrit Puzari*



Cite This: *Langmuir* 2024, 40, 21957–21975



Read Online

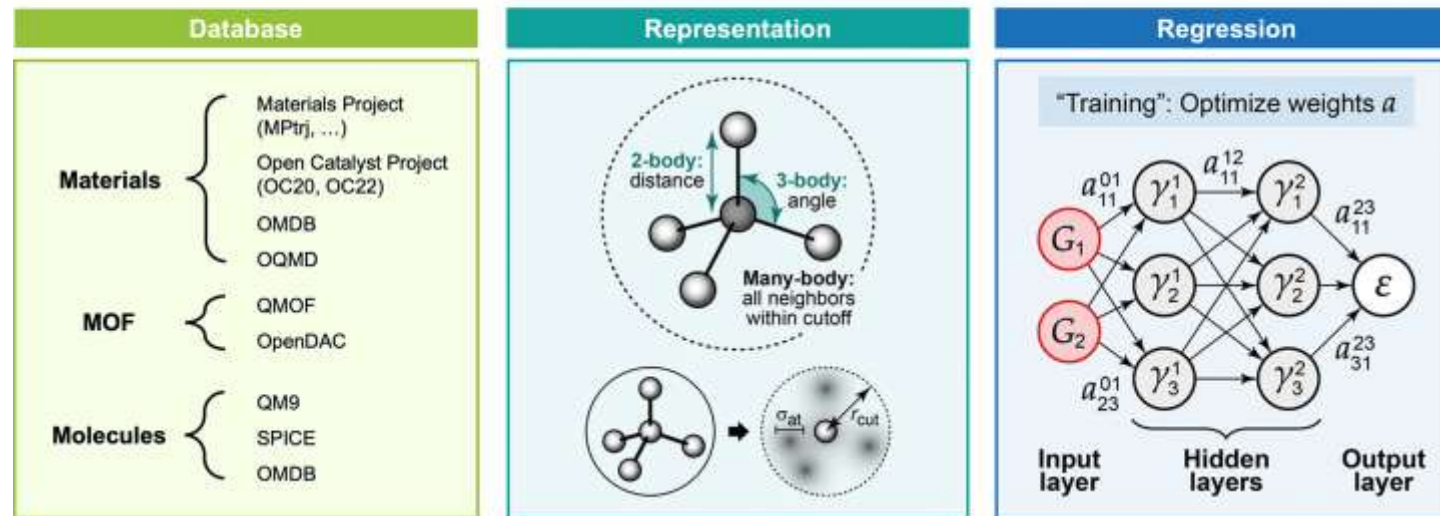
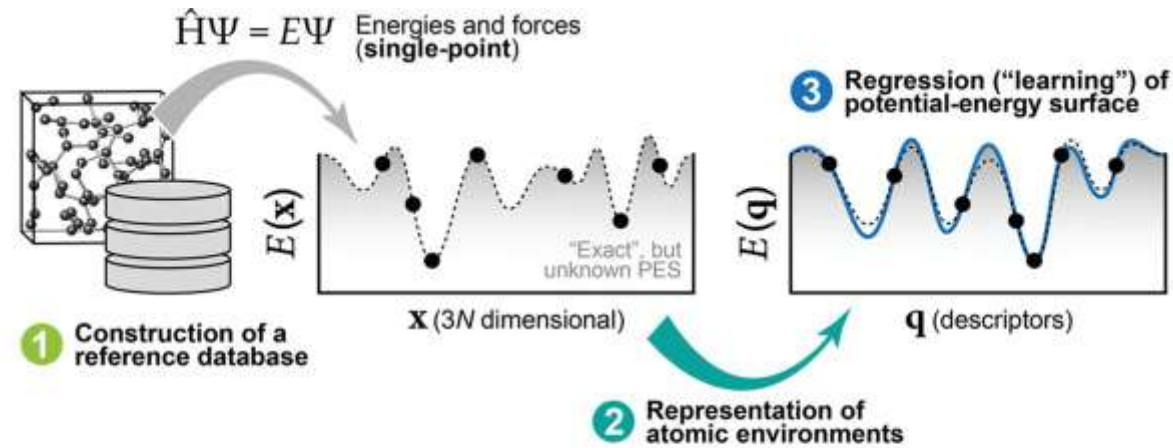




Preeti

6. Machine-learning for Property Prediction

ML for Property Prediction



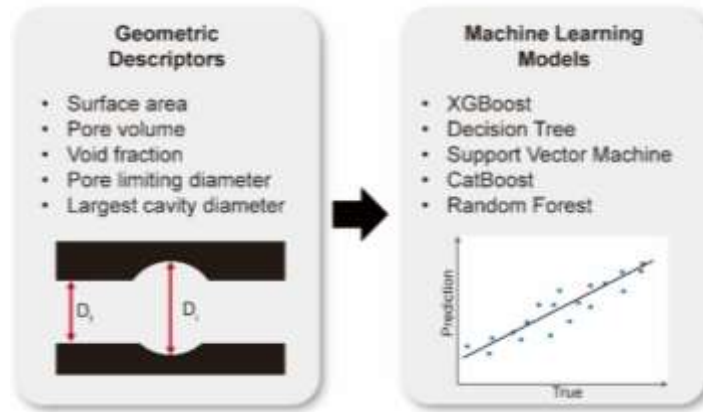
Data-driven approaches

ML for Property Prediction

Input representation/Features: Capturing Material Insights

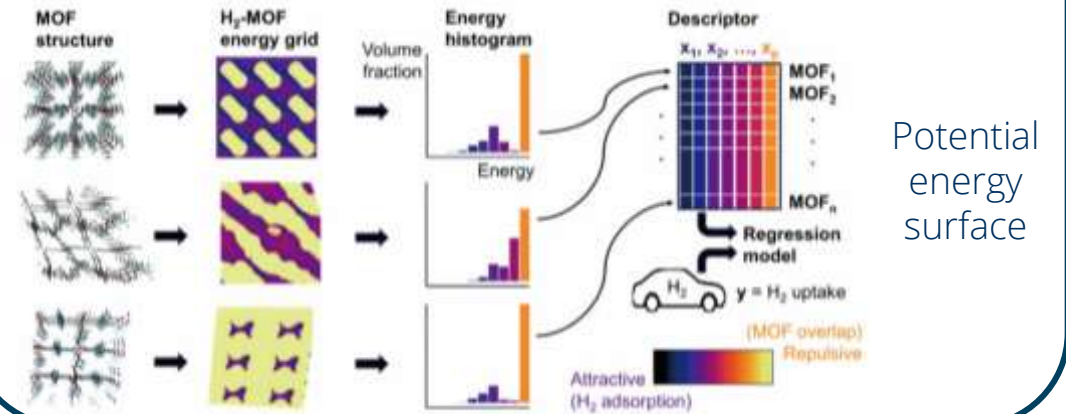
Geometric Descriptors

Capture **structural** information of materials.

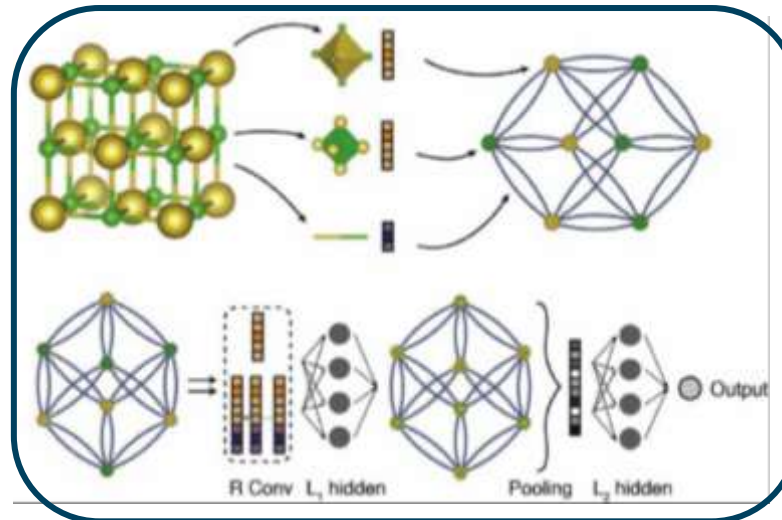


Chemical Descriptors

Capture **composition-related** information

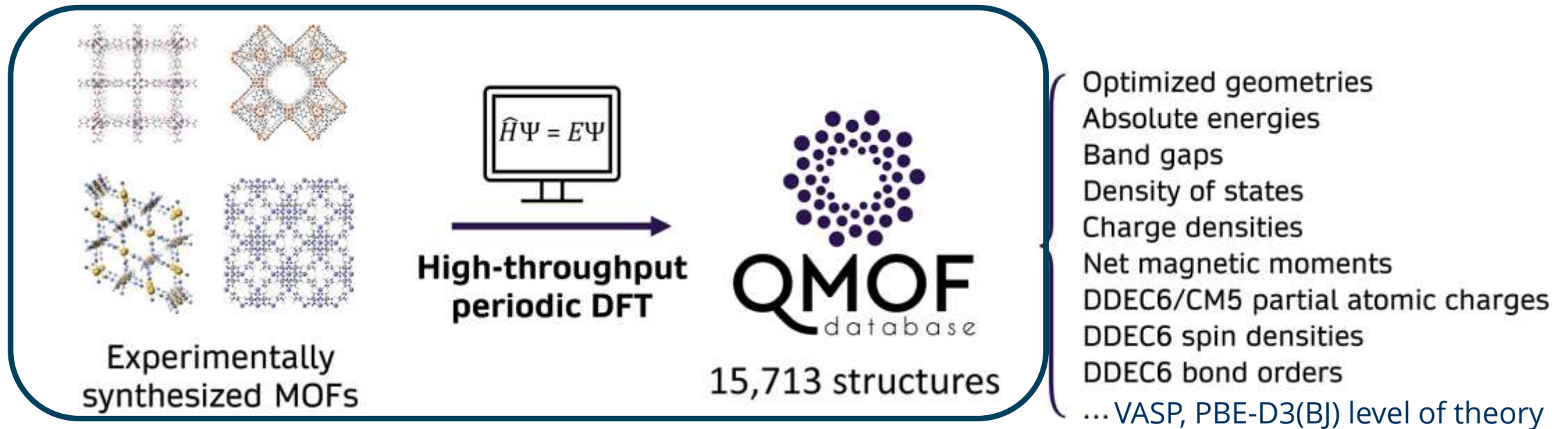


Quantum and Graph Based Descriptors



- SOAP (Smooth Overlap of Atomic Positions)
- ✓ Choosing the right descriptors is **essential** for building accurate ML models in material science.

Machine Learning Models for Band Gap Prediction



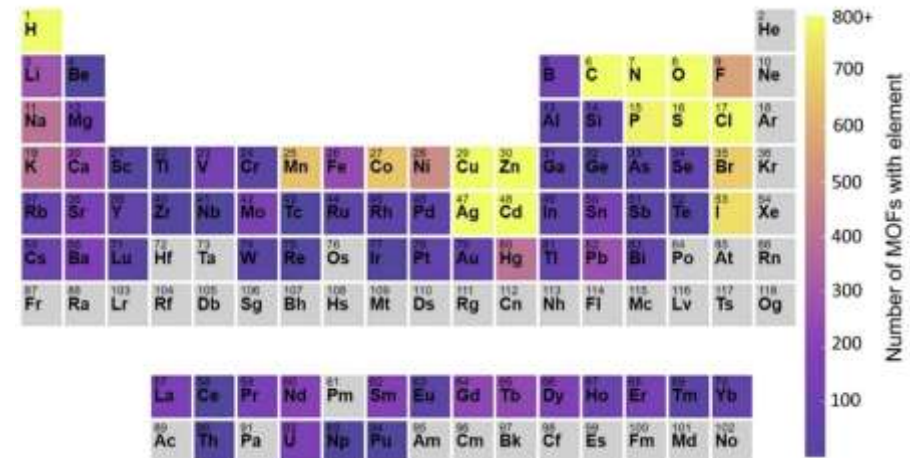
Matter



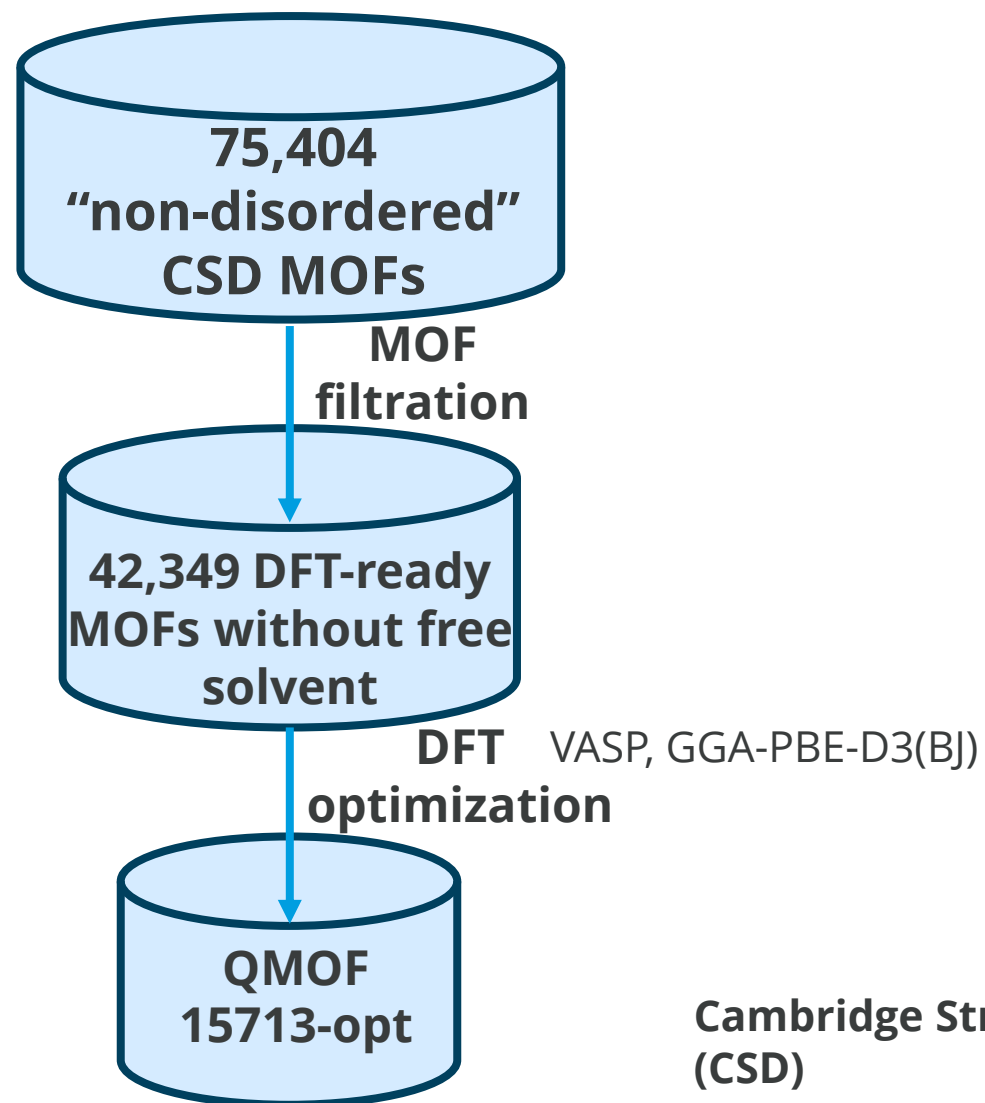
Article

Machine learning the quantum-chemical properties of metal–organic frameworks for accelerated materials discovery

Andrew S. Rosen, Shaelyn M. Iyer, Debmalya Ray, ..., Laura Gagliardi, Justin M. Notestein, Randall Q. Snurr

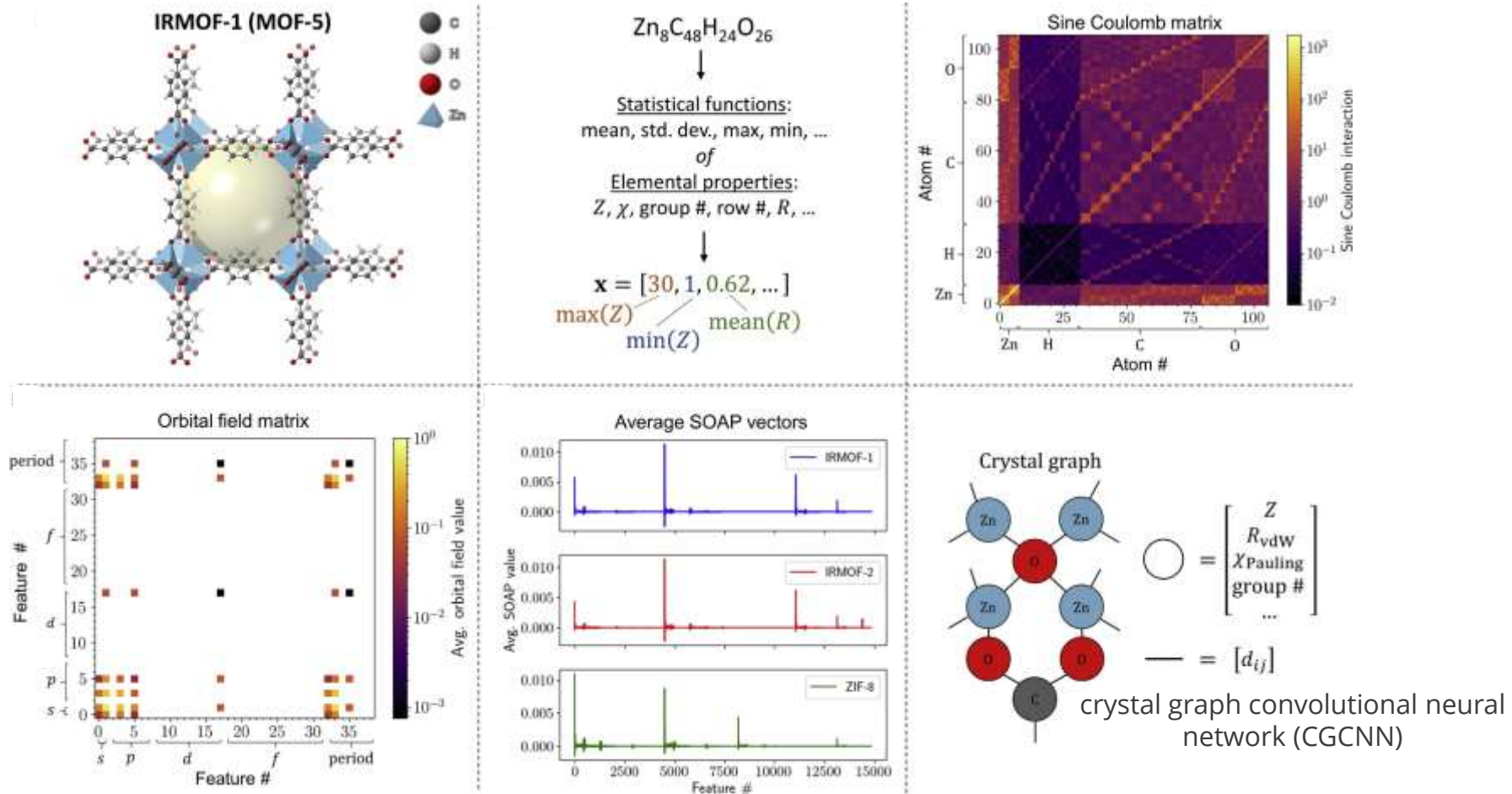


Workflow for Generating the Dataset



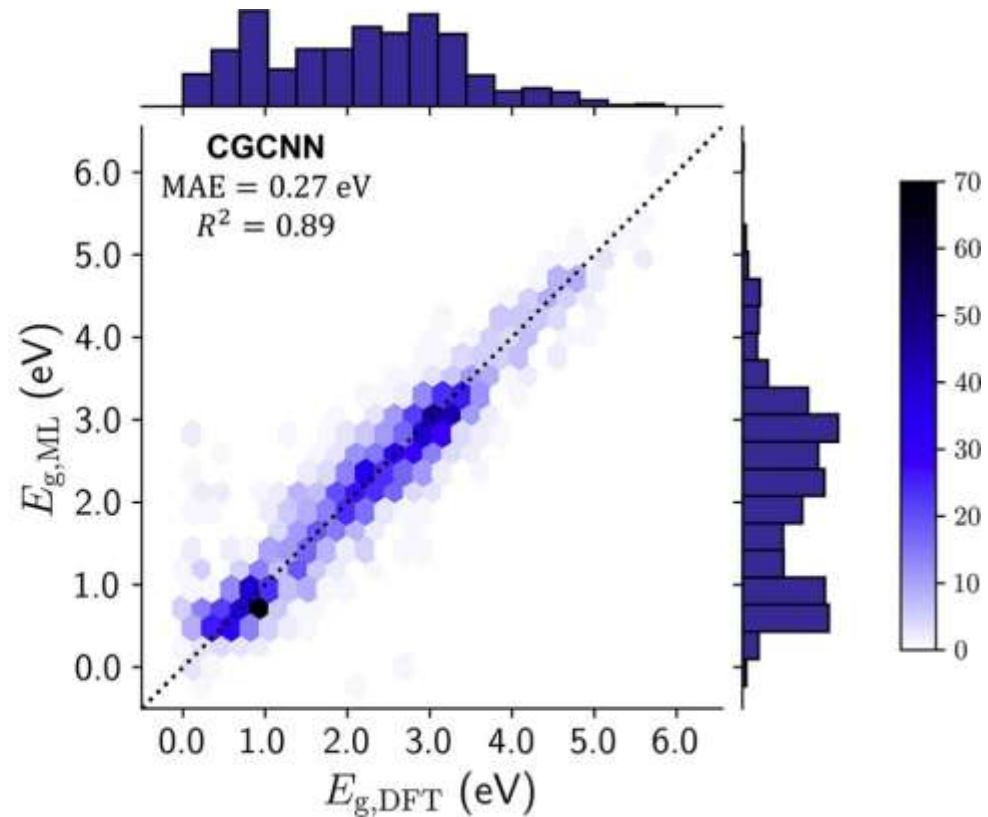
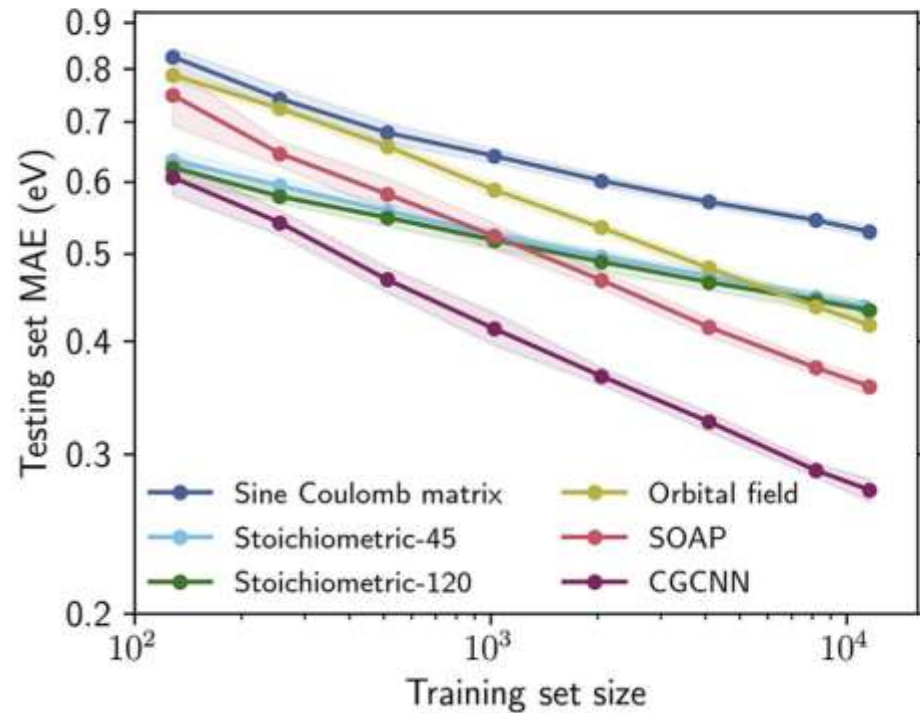
Cambridge Structural Database
(CSD)

Applied Featurization Methods



- ML models are trained on the band gaps of the DFT-optimized structures but take representations of the corresponding unrelaxed experimental structures as the input.

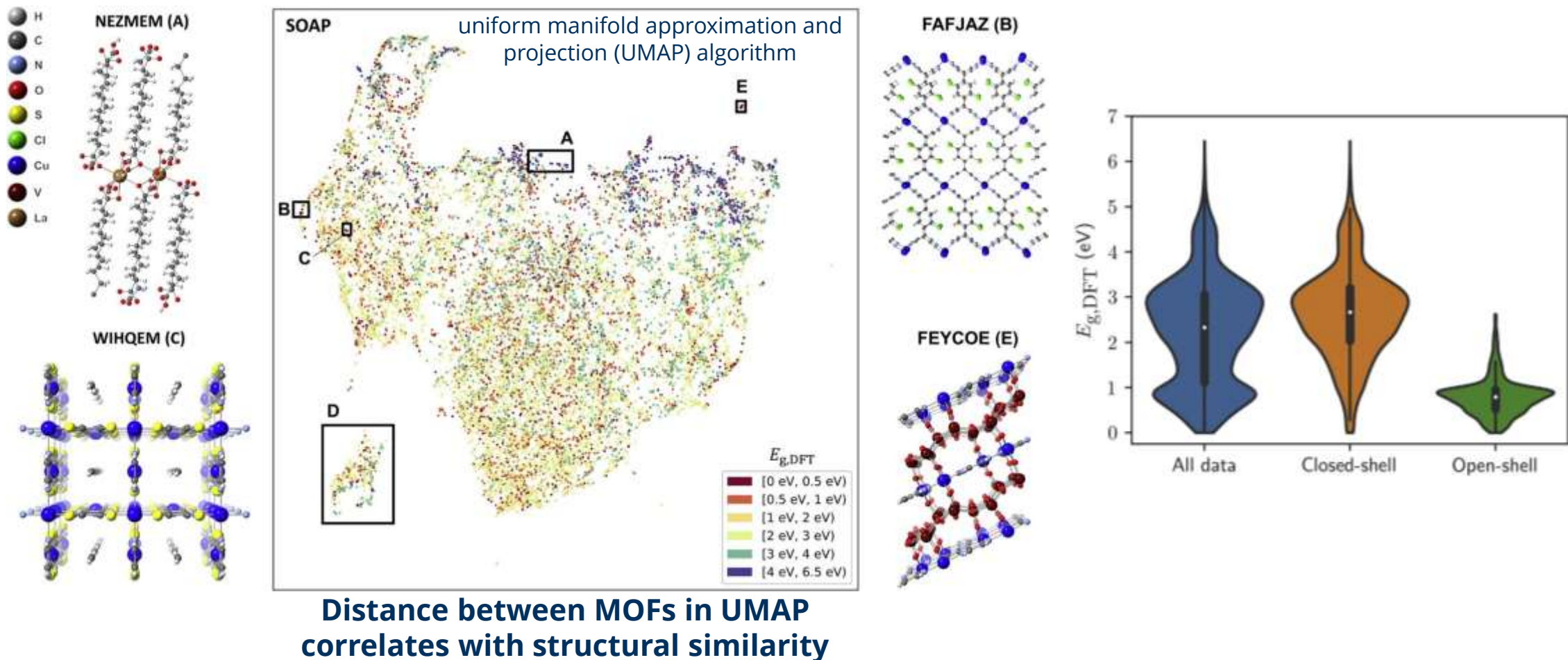
Training and Testing



Testing-set parity plot for the CGCNN

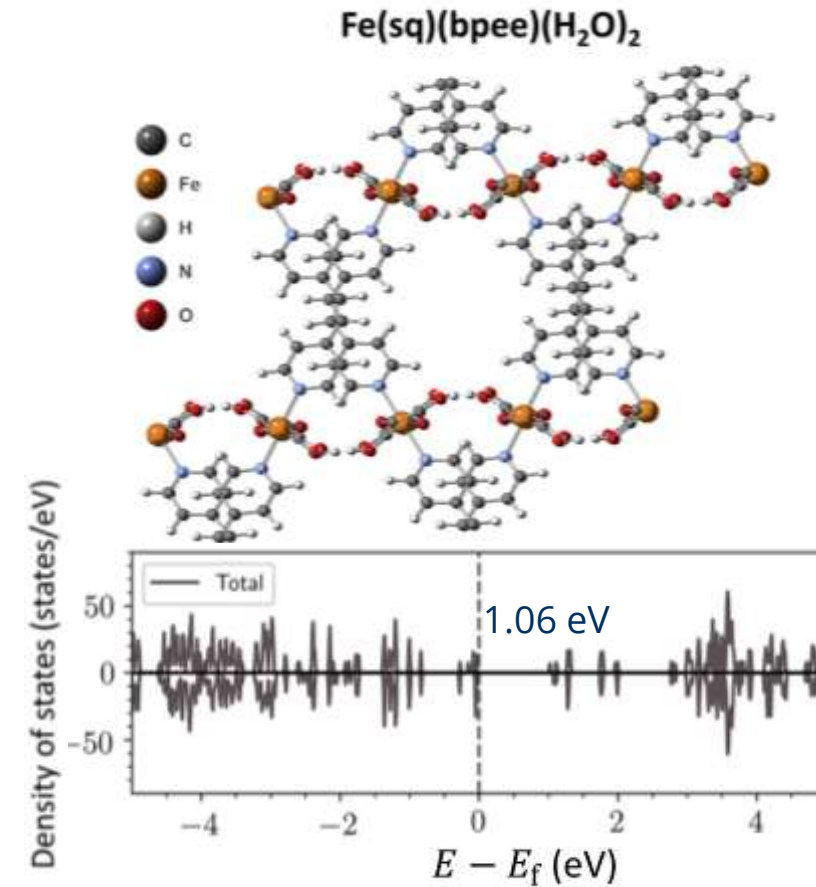
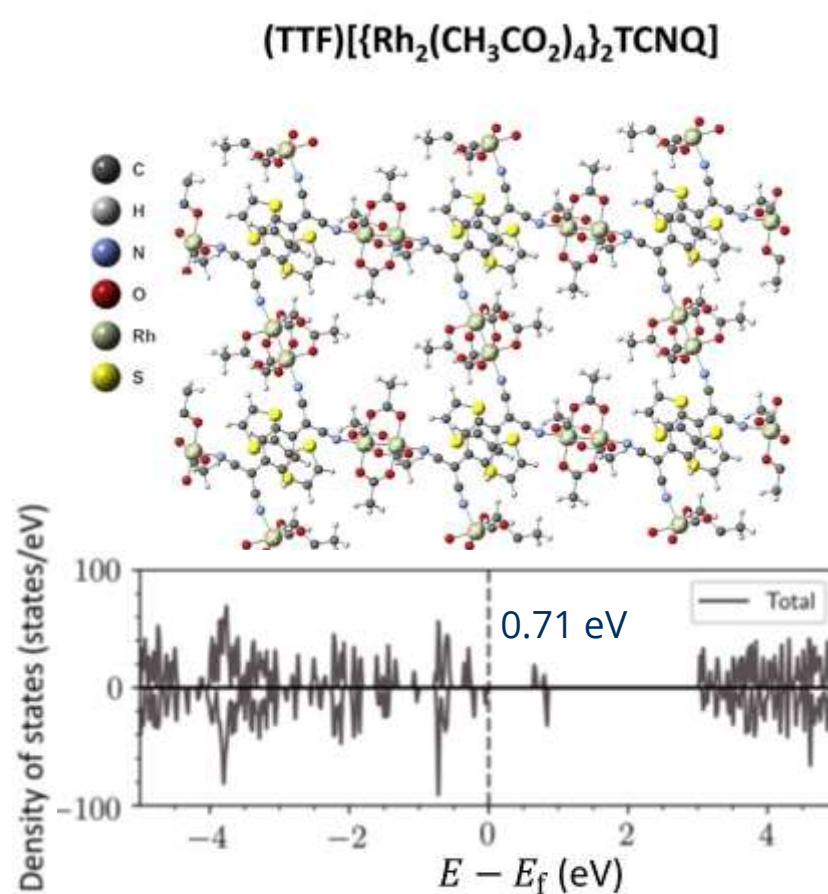
- CGCNN consistently achieves the lowest Mean Absolute Error (MAE), regardless of training set size.
- SOAP improves significantly with larger training sets.

Unsupervised Structural Dimensionality Reduction



- SOAP-based UMAP prioritizes **organic linker similarity** over metal identity

Identifying MOFs with low band gaps for potential (semi)conducting applications



- CGCNN model screened **42,349 MOFs** for low band gaps.
- Fe-based MOFs show promising low band gaps properties.
- Introduction of radical linkers enhances conductivity.

HSE06-D3(BJ)

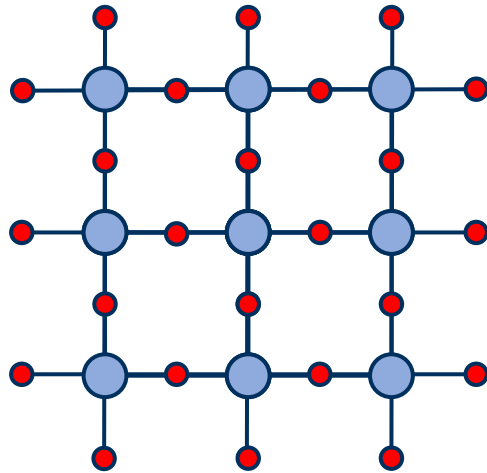


Shubhajit

7. Inverse Design

DESIGN SPACE OF MOFS

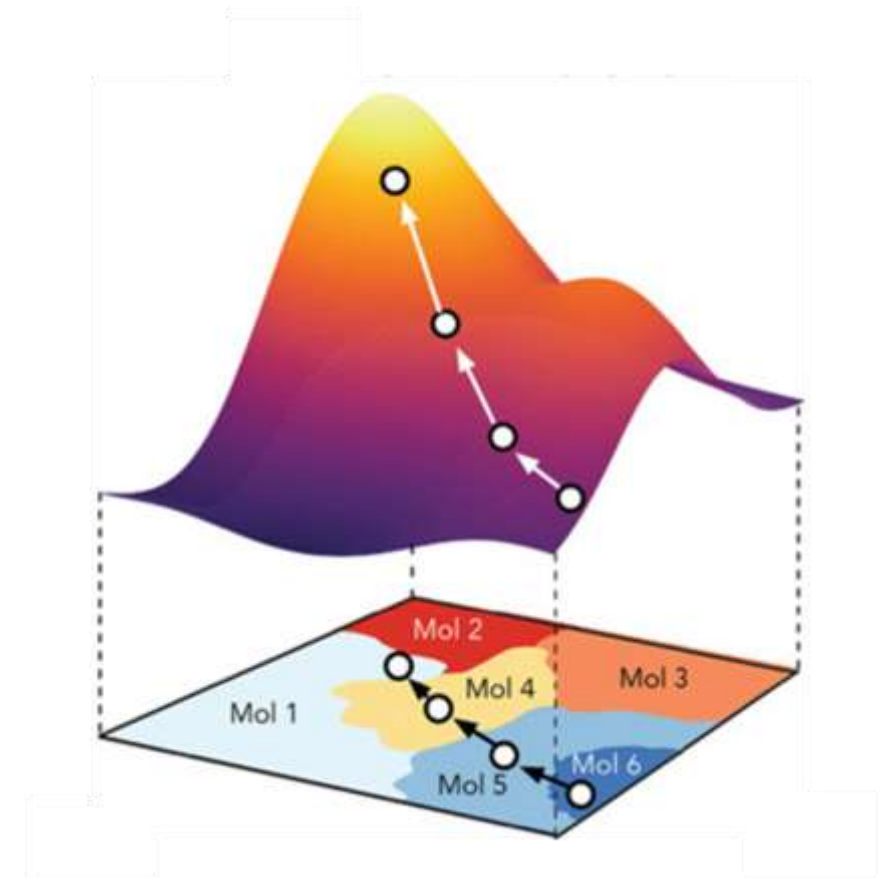
Enormous combinatorial space of building blocks



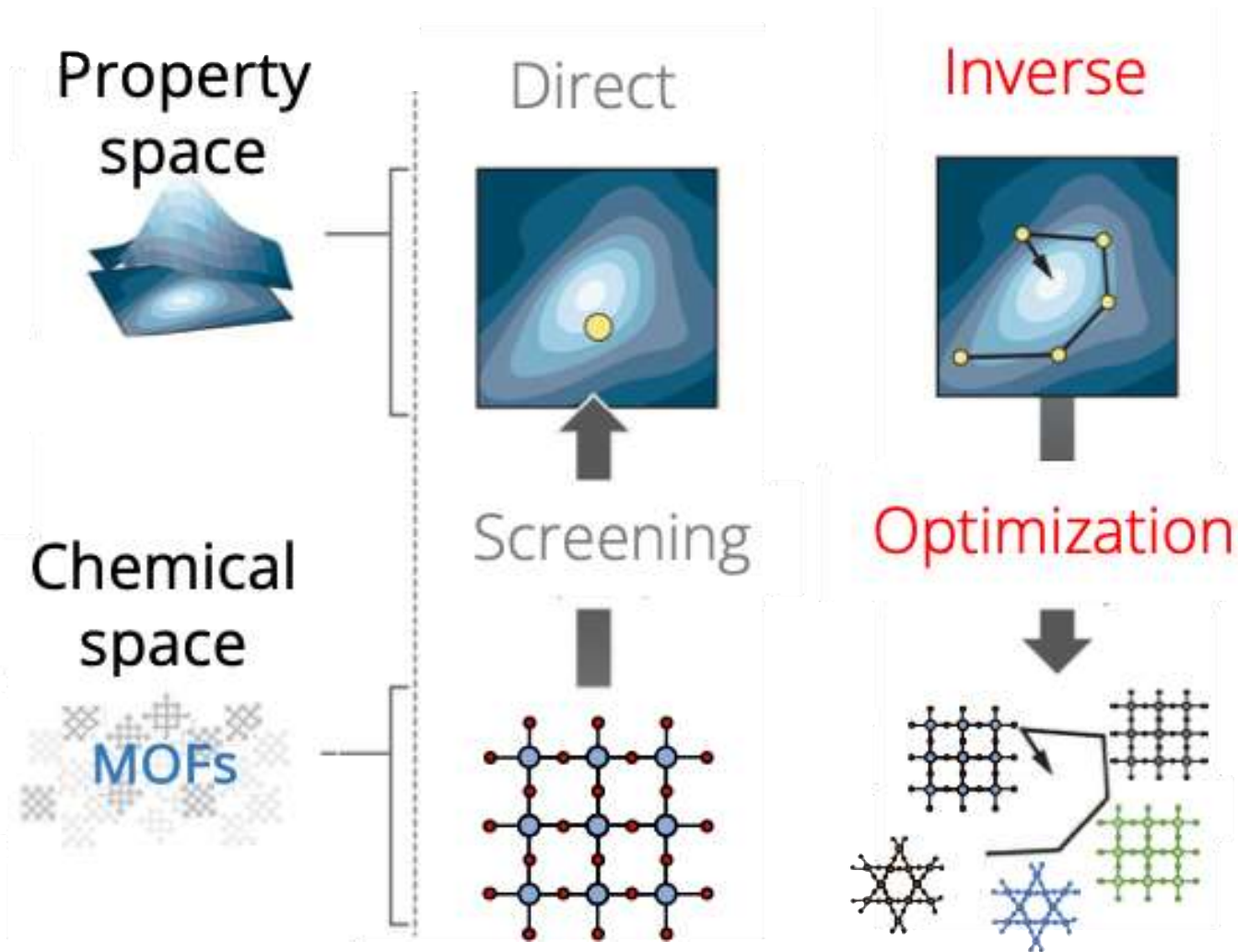
- Metal
- Linker
- Topology

1 H Hydrogen 1.008																	2 He Helium 4.003
3 Li Lithium 6.941	4 Be Beryllium 9.012																
11 Na Sodium 22.990	12 Mg Magnesium 24.305																
19 K Potassium 39.098	20 Ca Calcium 40.078																
37 Rb Rubidium 85.468	38 Sr Strontium 87.62																
55 Cs Cesium 132.905	56 Ba Barium 137.327																
87 Fr Francium [223]	88 Ra Radium [226]																

Complex – trade-off between properties (activity vs. stability)



CHEMICAL SPACE EXPLORATION



Brute-force screening

often **impractical** for large and complex chemical spaces

Inverse design

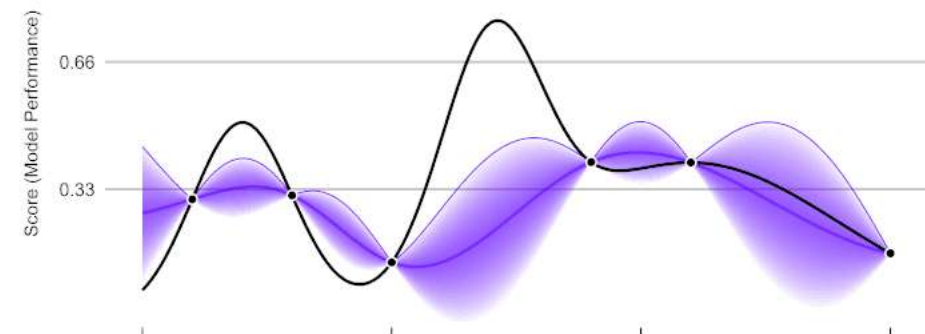
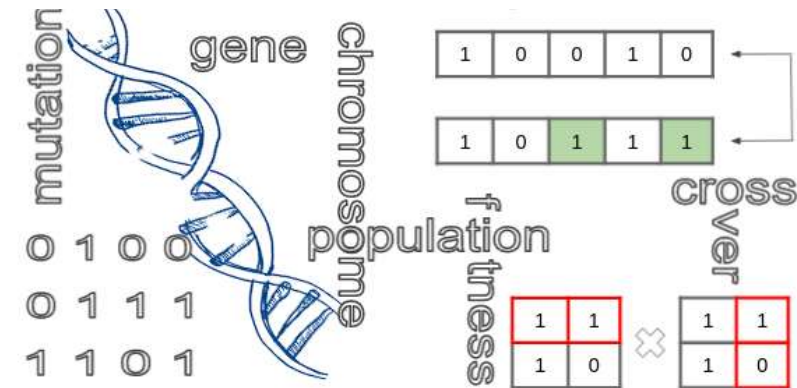
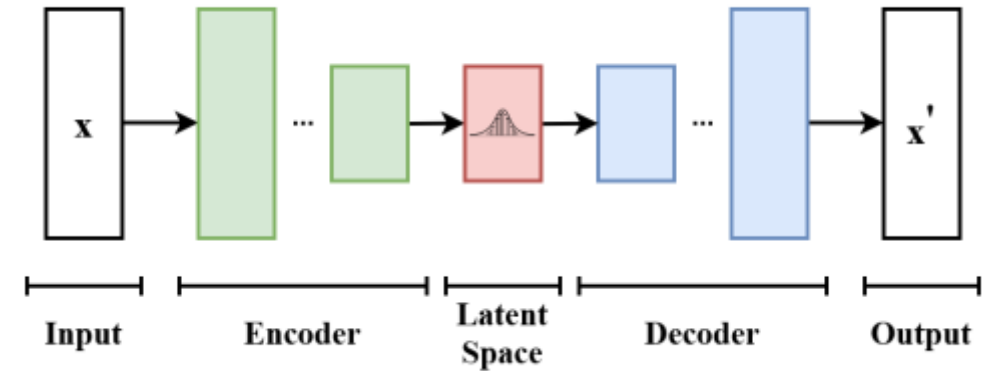
start with **target properties** and generate **best candidate** using an **optimization algorithm**

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Gradient-based
Variational auto-encoders (VAEs)

Heuristic approaches
Evolutionary algorithms

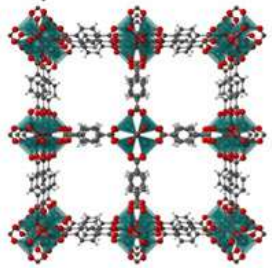
Uncertainty-based
Bayesian optimization



VAEs for Molecular Structures

VAEs use NN to learn patterns in molecular structures from existing data and predict new structures after training

Input structure



e.g.,
IRMOF-1

Deconstructor

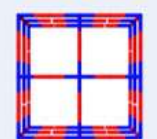
edge



vertex
metal



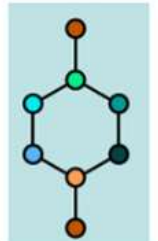
topo-
logy



LCD: 15.1 Å, PLD: 7.9 Å ...

Computed properties

RFcode

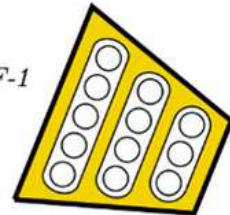


Zn₆_1|pcu

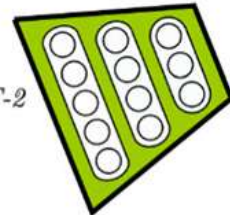
Latent space

Encoder

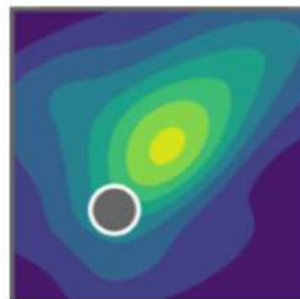
x_{RF-1}



x_{RF-2}



Z



Decoder

$\tilde{x}_{RF-1}$



$\tilde{x}_{RF-2}$

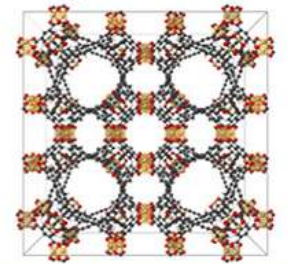


$\tilde{y}_{property}$



Use gradients in the latent space to locate minima or maxima

Output structure



Reconstructor

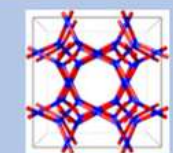
edge



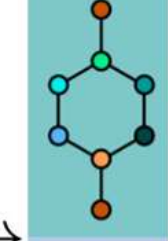
vertex
metal



topo-
logy



RFcode



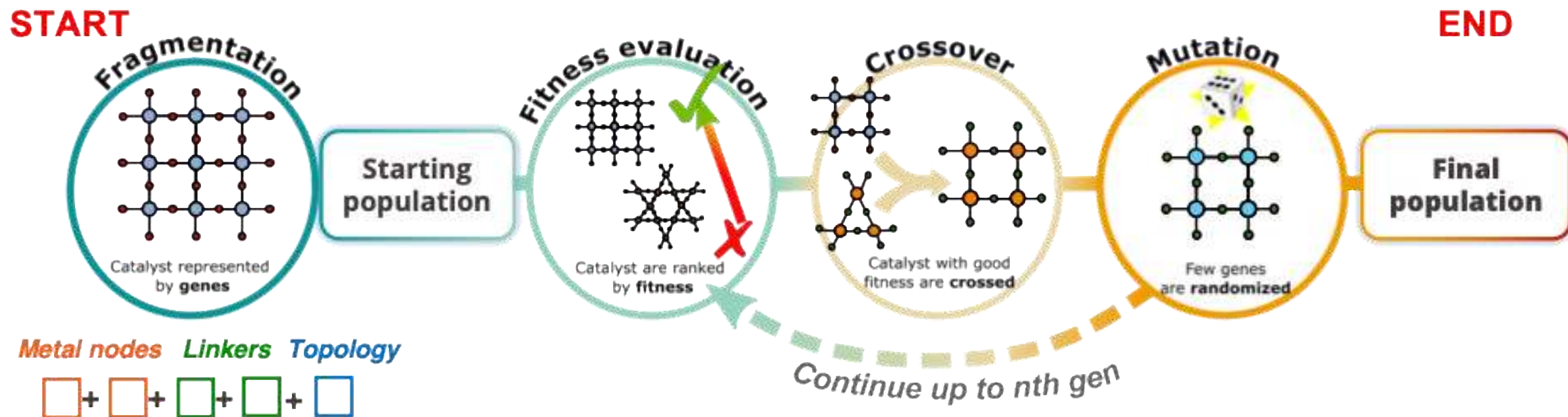
Cu₄_2|tbo

LCD: 19.1 Å, PLD: 8.6 Å ...

Predicted properties

GO mimics the concept of "survival of the fittest" from the Darwinian evolution

GO mimics the concept of "survival of the fittest" from the Darwinian evolution

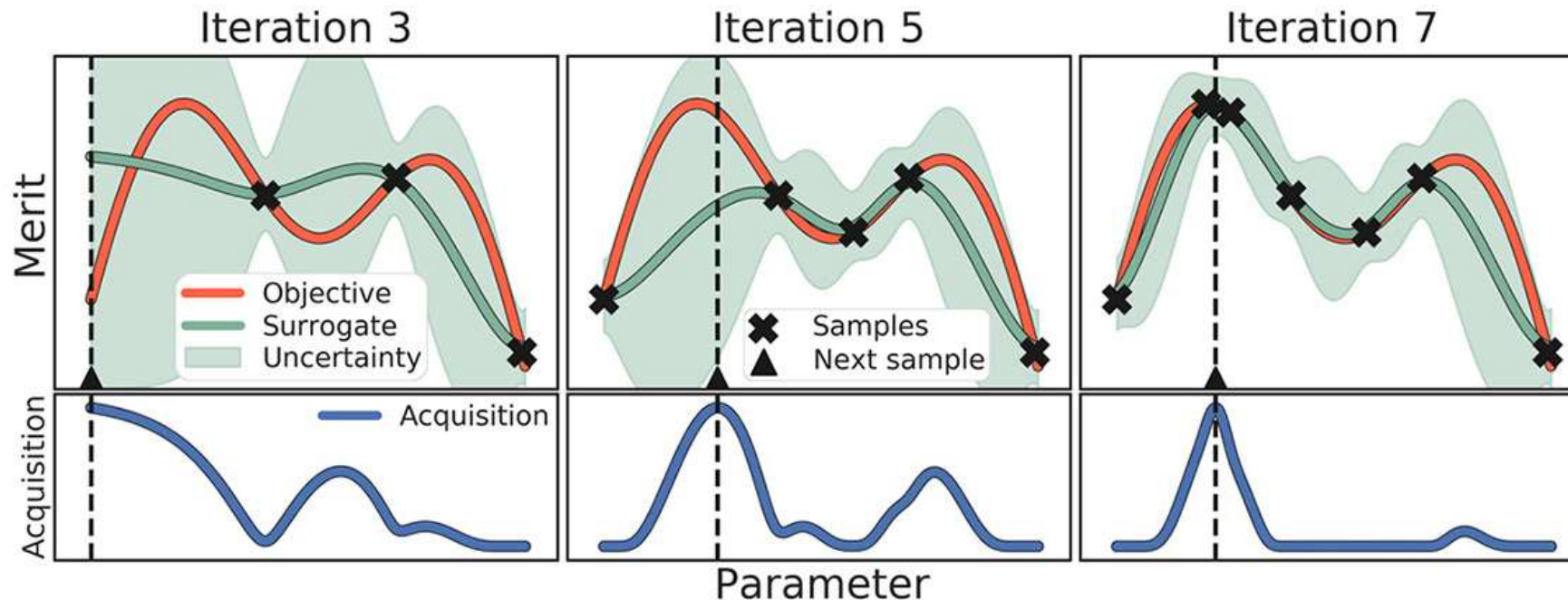


Lead candidates obtained exploring substantially less property evaluations

Bayesian Optimization (BO)

BO uses an **ML model** to **learn from previous observation** before suggesting next iteration

```
Result: Optimize objective function,  $f(x)$ 
while  $i < budget$  do
    build surrogate model ;
    compute acquisition function,  $a(x)$  ;
     $x_{next} \leftarrow \text{maximize } a(x)$  ;
    evaluate  $f(x_{next})$  ;
     $i \leftarrow i + 1$ ;
end
```

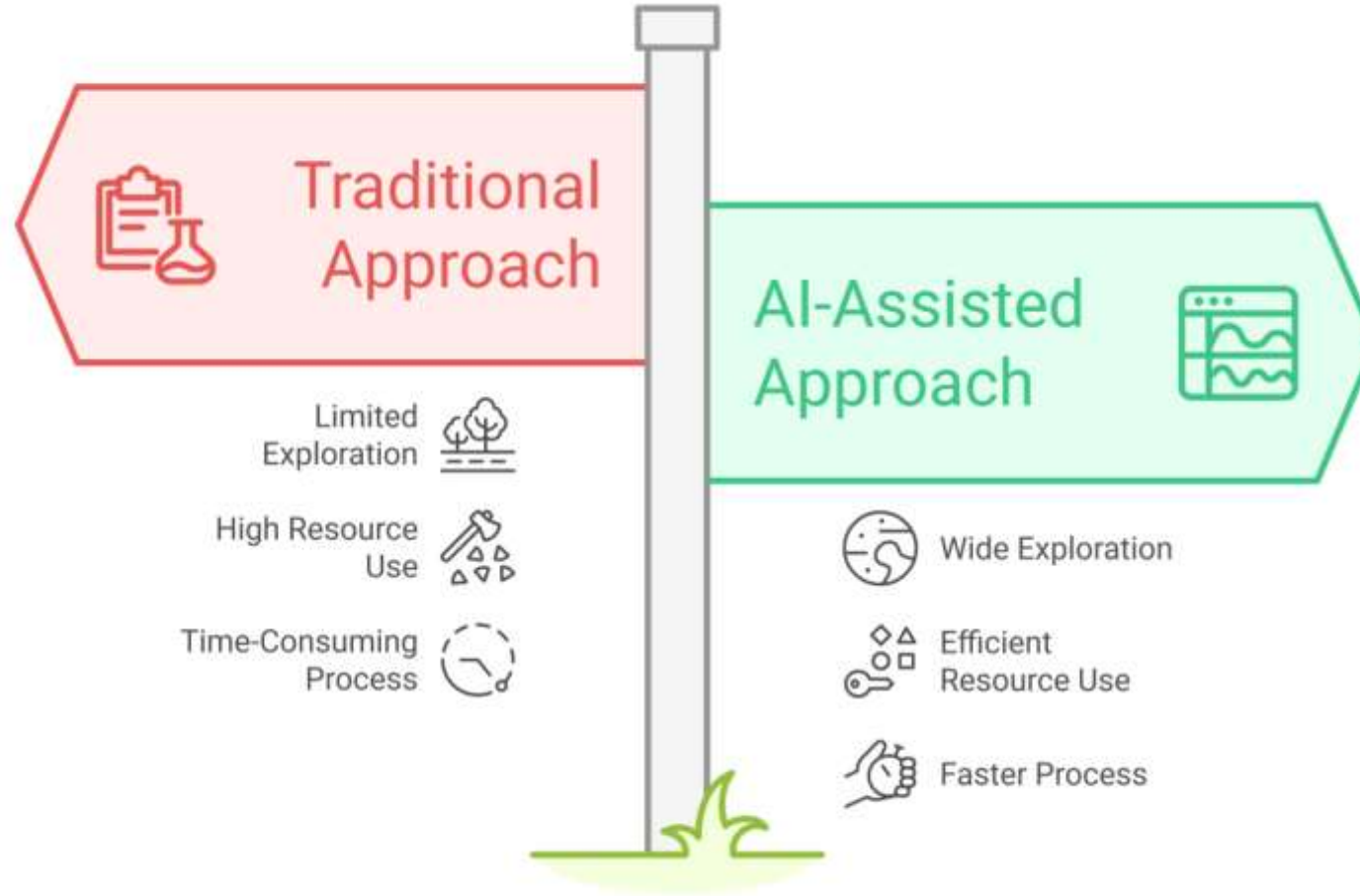




Maria

8. Case Studies of AI-driven Materials Design

Material Design



MatterGen (Microsoft)

Article

A generative model for inorganic materials design

<https://doi.org/10.1038/s41586-025-08628-5>

Received: 17 January 2024

Accepted: 10 January 2025

Published online: 16 January 2025

Open access

 Check for updates

Claudio Zeni^{1,2}, Robert Pinsler^{1,2}, Daniel Zügner^{2,3}, Andrew Fowler^{1,2}, Matthew Horton^{2,3}, Xiang Fu¹, Zilong Wang⁴, Aliaksandra Shysheya¹, Jonathan Crabbé¹, Shoko Ueda¹, Roberto Sordillo¹, Lixin Sun¹, Jake Smith², Bichlien Nguyen², Hannes Schulz², Sarah Lewis¹, Chin-Wei Huang⁵, Ziheng Lu⁶, Yichi Zhou², Han Yang⁶, Hongxia Hao⁶, Jielan Li⁶, Chunlei Yang⁴, Wenjie Li⁴, Ryota Tomioka^{1,2,3} & Tian Xie^{1,2,3}

The design of functional materials with desired properties is essential in driving technological advances in areas such as energy storage, catalysis and carbon capture^{1–3}. Generative models accelerate materials design by directly generating new materials

doi: g84w9p

GNoME (DeepMind - Alphabet Inc.)

Article

Scaling deep learning for materials discovery

<https://doi.org/10.1038/s41586-023-06735-9>

Received: 8 May 2023

Accepted: 10 October 2023

Published online: 29 November 2023

Open access

 Check for updates

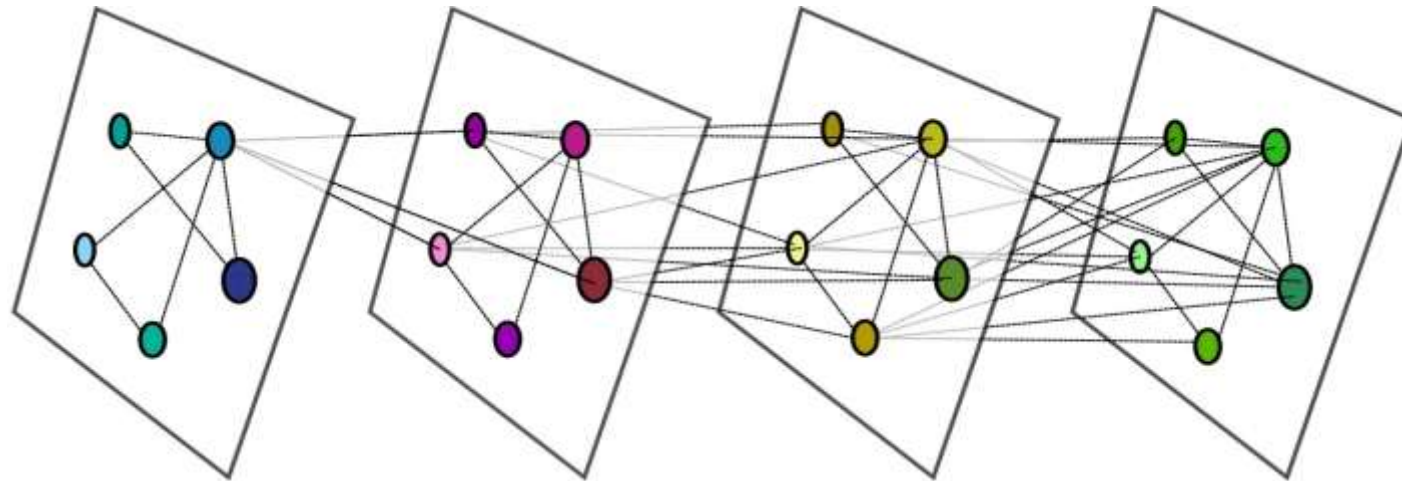
Amil Merchant^{1,2,3}, Simon Batzner^{1,2}, Samuel S. Schoenholz^{1,2}, Muratahan Aykol¹, Gwooon Cheon² & Ekin Dogus Cubuk^{1,2,3}

Novel functional materials enable fundamental breakthroughs across technological applications from clean energy to information processing^{1–11}. From microchips to batteries and photovoltaics, discovery of inorganic crystals has been bottlenecked by expensive trial-and-error approaches. Concurrently, deep-learning models for language, vision and biology have showcased emergent predictive capabilities with increasing data and computation^{12–14}. Here we show that graph networks trained at

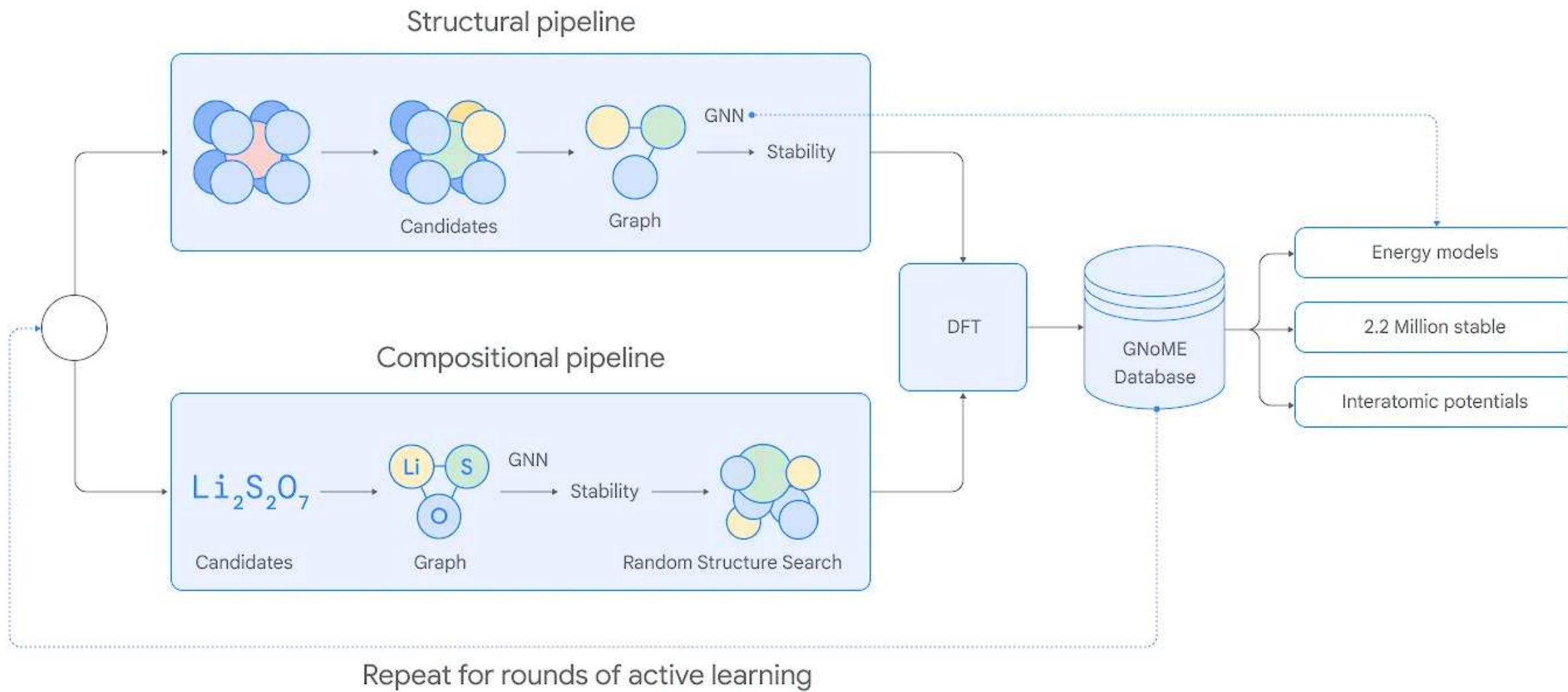
doi: gs7sbc

What is GNoME?

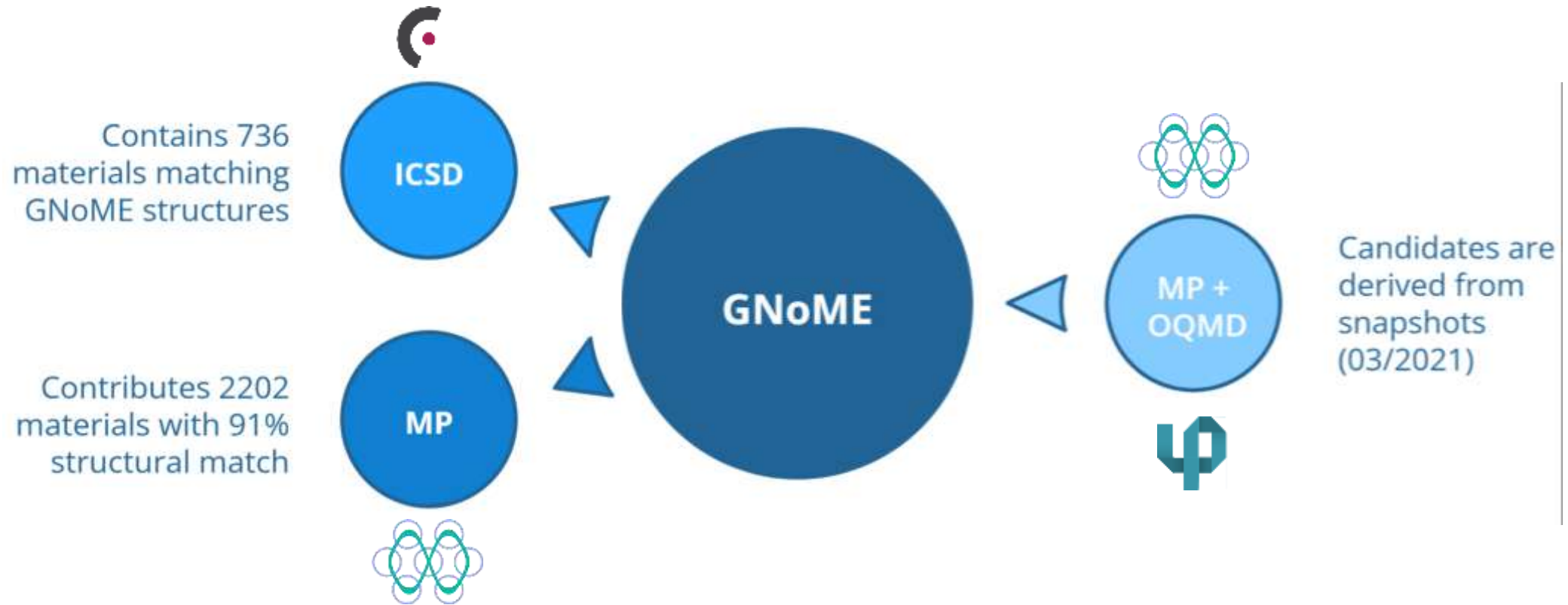
- **GNoME** = **G**raph **N**etworks **f**or **M**aterials **E**xploration
- uses **GNNs** to predict stable crystal structures



GNoME - How does it work?



GNoME – Experimental validation



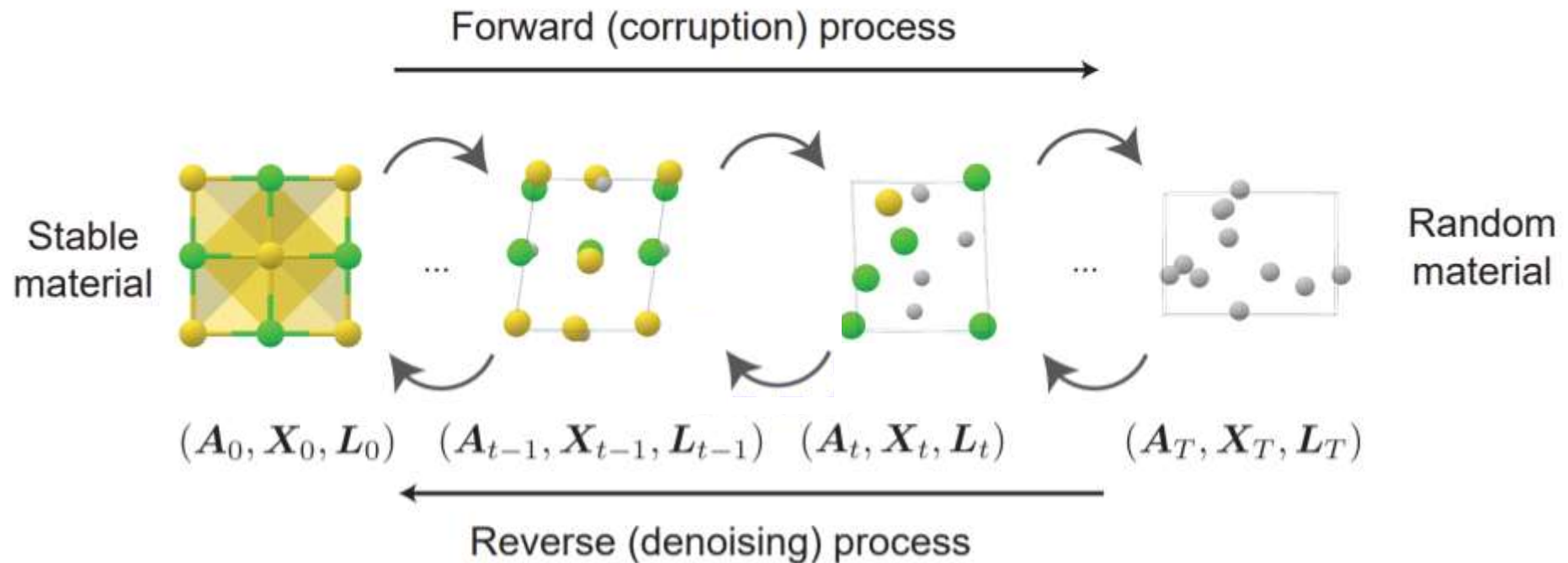
What is MatterGen?

- **diffusion-based** generative model designed to create stable, novel inorganic materials



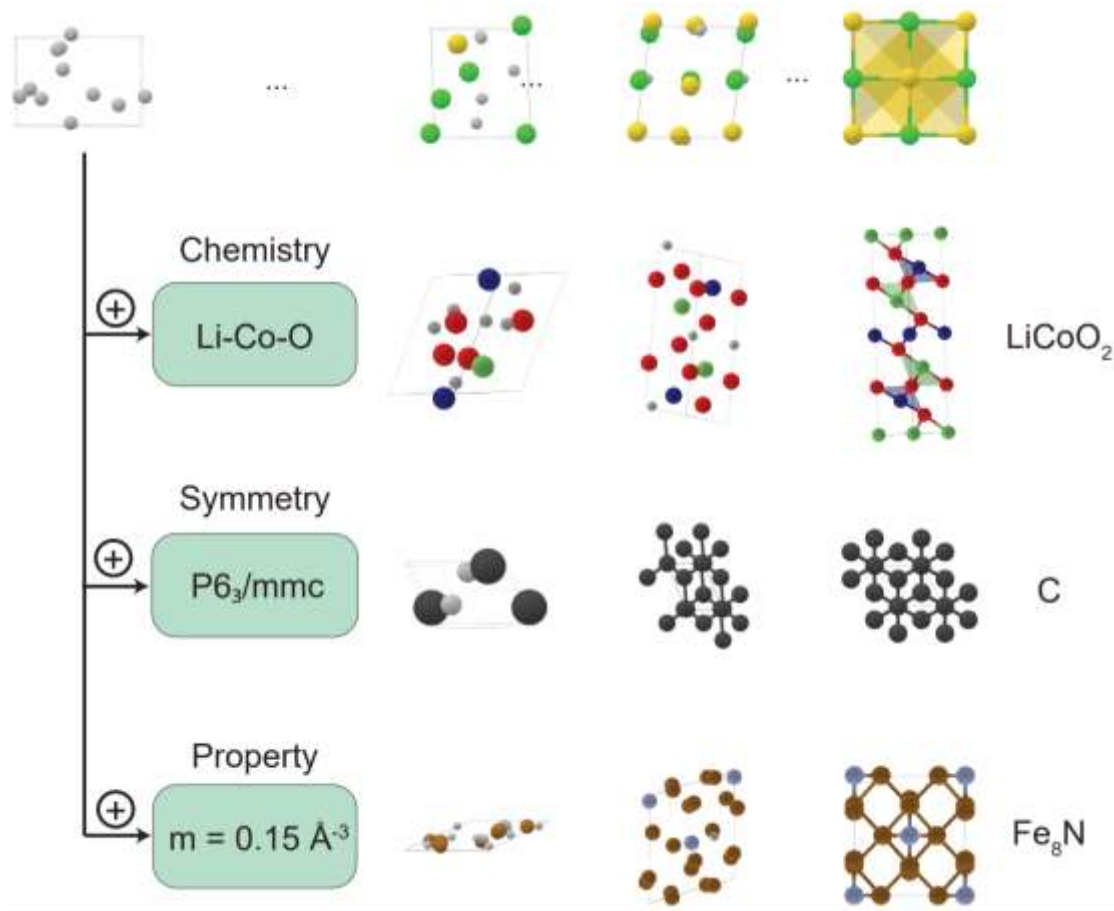
MatterGen - How does it work?

- reverses a **corruption** process through iteratively **denoising** an initially random structure (atom types A , coordinates X , and the lattice L)



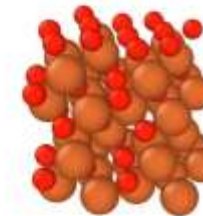
MatterGen - How does it work?

- fine-tuned model generates materials with desired **chemistry**, **symmetry** or **scalar property** constraints



To property-guided Materials Design

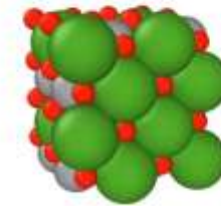
High Magnetic
Density



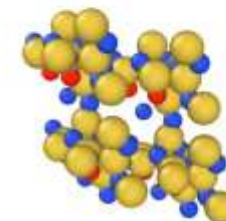
High Bulk
Modulus



Specific
Chemistry

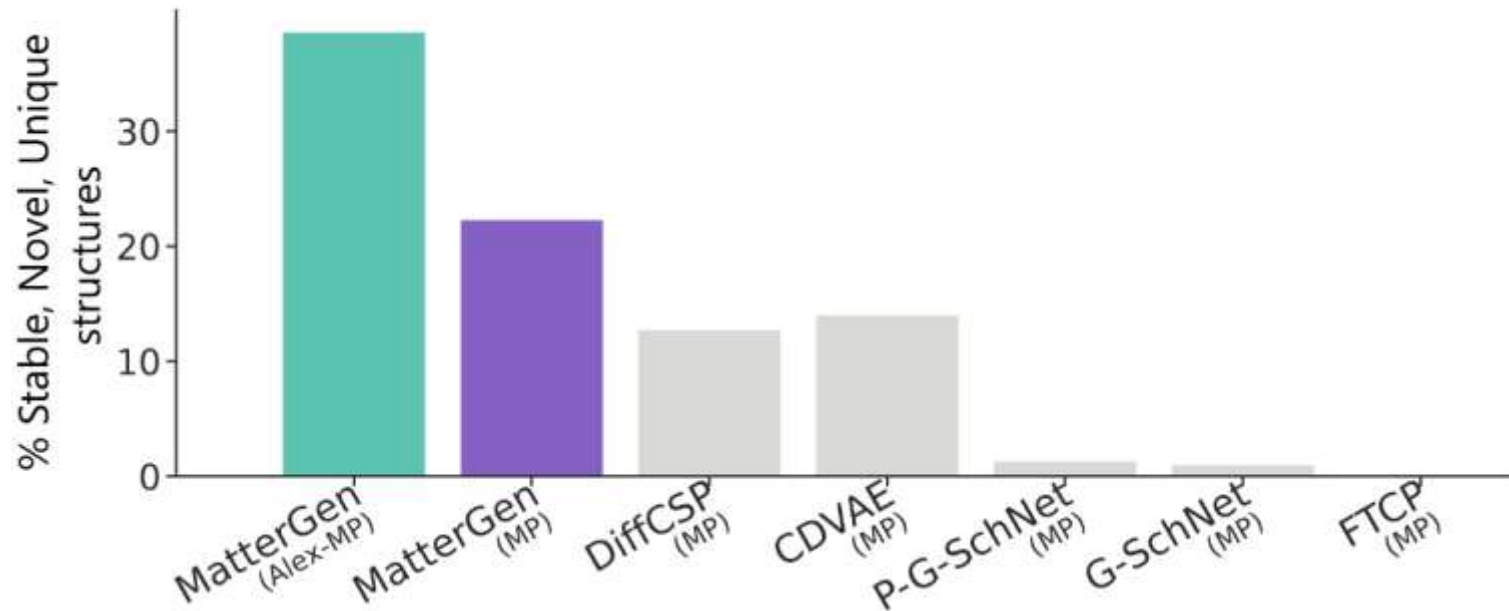


Specific
Band Gap



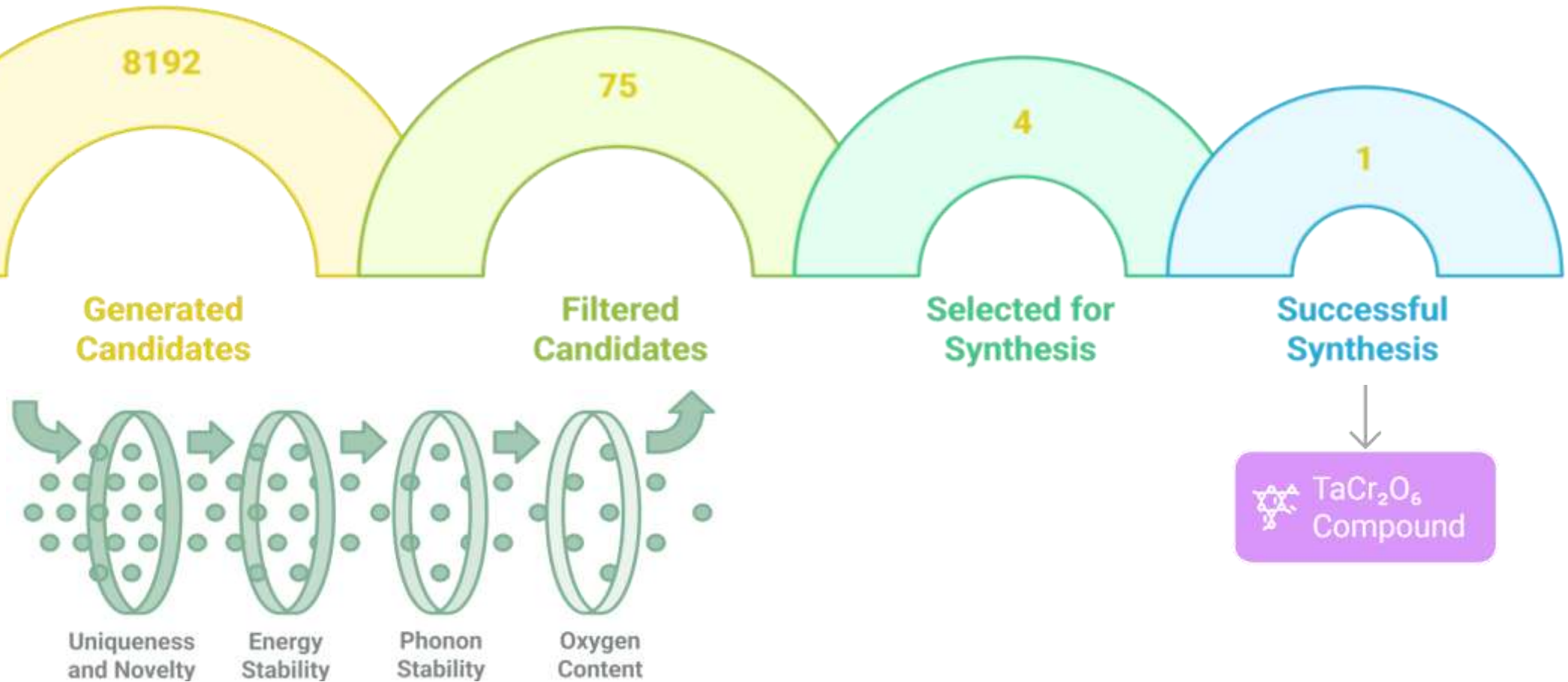
MatterGen - How does it work?

- Model is trained on 608,000 stable materials from:
 - the Materials Project (MP) database
 - Alexandria (Alex) database



MatterGen - Experimental verification

- Experimental synthesis in addition to computational evaluation



Thank you for your attention

